LIMITS

SYNOPSIS

Neighbourhood of a real number:

- Let 'a' be a real number and δ be a positive real number, then (a-δ,a+δ) is called δ-neighbourhood of a. The interval (a-δ,a) is called the left δ- neighbourhood of 'a' and the interval (a,a+δ) is called the right δ-neighbourhood of a. The deleted δ- neighbourhood of 'a' is denoted by (a-δ,a) $\cup$ (a,a+δ).

Limit of a function:

- Let f be a function defined over a deleted neighbourhood of the real number 'a' and 'l' be a real number. If to every positive number $\in$ (however small) there exists a positive number 'δ' such that $|f(x)-l|<\in$ for all x such that $0<|x-a|<\delta$, we say that $f(x)$ tends to 'l' as x tends to a and we write it as $\underset{x\to a}{Lt}\ f(x)=l$.

Right handed limit:

- Let $f(x)$ be a function defined in the interval $(a,a+h)$ and 'l' be a real number. If to every positive number $\in$ (however small), there corresponds a positive number δ such that $|f(x)-l|<\in, \forall\ x\in(a,a+\delta)$ then we say that $f(x)$ tends to 'l' as x approaches a through values higher than a and we denote

$$\underset{x\to a^+}{Lt}\ f(x)=l\ (or)\ \underset{h\to 0}{Lt}\ f(a+h)=l.$$

Left handed limit :

- Let $f(x)$ be a function defined in the interval $(a-h,a)$ and 'l' be a real number. If to every positive number $\in$ (however small), there corresponds a positive number δ such that $|f(x)-l|<\in$ for all x such that $x\in(a-\delta,a)$ we say that $f(x)$ tends to 'l' as x approaches a through values less than a and we denote it as

$$\lim_{x\to a^-} f(x)=l\ (or)\ \lim_{h\to 0} f(a-h)=l$$

- Let $a,l\in R$ and f be a function defined on a deleted neighbourhood of 'a' then

$$\lim_{x\to a} f(x)=l \Leftrightarrow \lim_{x\to a^+} f(x)=l=\lim_{x\to a^-} f(x)$$

- If $\lim_{x\to a} f(x)$ exists, then

$$\lim_{x\to a} f(x)=\lim_{x\to 0} f(a+x)=\lim_{x\to 0} f(a-x)$$

Infinite limits :

- Let f be a function defined over a deleted neighbourhood of a. If for every positive number k (however large) there corresponds a positive number δ, such that $f(x)>k$, $\forall x$ such that $0<|x-a|<\delta$, we say that $f(x)\to+\infty$ as $x\to a$. We write is as $\underset{x\to a}{Lt}\ f(x)=+\infty$

Similarly we define (i) $\underset{x\to a}{Lt}\ f(x)=-\infty$

(ii) $\underset{x\to+\infty}{Lt}\ f(x)=+\infty,\ \underset{x\to+\infty}{Lt}\ f(x)=-\infty$

(iii) $\underset{x\to-\infty}{Lt}\ f(x)=+\infty,\ \underset{x\to-\infty}{Lt}\ f(x)=-\infty$

Limit of a function f as $x\to+\infty$ or $-\infty$:

- Let f be a function and l be a real number. If for every positive number $\in$ there corresponds a positive number k (however large) such that $|f(x)-l|<\in \forall x>k$, then we say that $f(x)$ tends to l as x tends to ∞. We write it as $\underset{x\to+\infty}{Lt} f(x)=l.$ Similarly we define $\underset{x\to-\infty}{Lt} f(x)=l$.

Fundamental Theorem on Limits :

➢ If $\underset{x\to a}{Lt} f(x)=l \; and \; \underset{x\to a}{Lt} g(x)=m$ then,

i) $\underset{x\to a}{Lt}\{f(x)+g(x)\}=l+m$

ii) $\underset{x\to a}{Lt}\{f(x)-g(x)\}=l-m$

iii) $\underset{x\to a}{Lt}\{f(x).g(x)\}=l.m$

iv) $\underset{x\to a}{Lt}\{k.f(x)\}=k.\underset{x\to a}{Lt} f(x)$

v) $\underset{x\to a}{Lt}\dfrac{f(x)}{g(x)}=\dfrac{l}{m}$, if $m\neq 0$

vi) $\underset{x\to a}{Lt}(f(x))^k=l^k$, if $k\in Q$ and $l^k\in R$

vii) $\underset{x\to a}{Lt} f(g(x))=f\left(\underset{x\to a}{Lt}(g(x))\right)=f(m)$

viii) $\underset{x\to a}{Lt}\left(f(x)^{g(x)}\right)=l^m$, if $l^m\in R$

ix) If $f(x)\leq g(x)$ on a deleted nbd of 'a' then $\underset{x\to a}{Lt} f(x)\leq \underset{x\to a}{Lt} g(x)$

x) $\underset{x\to a}{Lt} f(x)=l \Rightarrow \underset{x\to a}{Lt}|f(x)|=|l|$.

However the converse need not be true

Ex. $\underset{x\to 0}{Lt}\dfrac{1}{|x|}=\infty$ where as $\underset{x\to 0}{Lt}\dfrac{1}{x}$ does not exist.

Indeterminate forms :

➢ $\dfrac{0}{0},\dfrac{\infty}{\infty},\infty-\infty,\ 0\times\infty, 0^0,\infty^0$ and 1^∞ are called indeterminate forms

Standard Limits :

➢ For all real values of n, $\underset{x\to a}{Lim}\dfrac{x^n-a^n}{x-a}=n.a^{n-1}$ (Provided $n\,a^{n-1}$ is defined)

➢ $\underset{x\to a}{Lt}\dfrac{x^m-a^m}{x^n-a^n}=\dfrac{m}{n}a^{m-n}$, $(m>n)$

➢ If $0<|x|<\dfrac{\pi}{2}$ and x is measured in radians.

$\underset{x\to 0}{Lim}\dfrac{\sin x}{x}=1$ and $\underset{x\to 0}{Lim}\dfrac{\tan x}{x}=1$,

$\underset{x\to 0}{Lt}\dfrac{\sin ax}{x}=a$ and $\underset{x\to 0}{Lt}\dfrac{\tan ax}{x}=a$

$\underset{x\to 0}{Lim}\dfrac{\sin x^0}{x}=\dfrac{\pi}{180}$ and $\underset{x\to 0}{Lim}\dfrac{\tan x^0}{x}=\dfrac{\pi}{180}$

➢ $\underset{x\to\infty}{Lim}\dfrac{\sin x}{x}=0$ and $\underset{x\to\infty}{Lim}\dfrac{\cos x}{x}=0$

➢ $\underset{x\to 0}{Lim}\dfrac{\sin^{-1}x}{x}=1$ and $\underset{x\to 0}{Lim}\dfrac{\tan^{-1}x}{x}=1$

➢ $\underset{x\to 0}{Lim}\left(\dfrac{e^x-1}{x}\right)=1$

➢ $\underset{x\to 0}{Lim}\left(\dfrac{a^x-1}{x}\right)=\log_e a,\ (a>0)$

➢ $\underset{x\to 0}{Lim}\dfrac{a^x-b^x}{x}=\log_e\left(\dfrac{a}{b}\right)$

➢ $\underset{x\to 0}{Lim}\dfrac{a^x-1}{b^x-1}=\log_b a$

➢ $\underset{x\to a}{\lim}\dfrac{|x-a|}{x-a}$ does not exist

➢ $\underset{x\to 0}{Lim}(1+x)^{\frac{1}{x}}=e$, $\underset{x\to 0}{Lt}(1+ax)^{\frac{1}{x}}=e^a$

➢ $\underset{x\to\infty}{Lim}\left(1+\dfrac{1}{x}\right)^x=e$, $\underset{x\to\infty}{Lt}\left(1+\dfrac{a}{x}\right)^x=e^a$

➢ $f(x), g(x)$ are two polynomials such that degree of $f(x)$ is m and degree of $g(x)$ is n then

i) $\lim_{x\to\infty}\frac{f(x)}{g(x)}=0$ for $m<n$

ii) $\lim_{x\to\infty}\frac{f(x)}{g(x)}=\infty$ for $m>n$ and coef of $x^m>0$

iii) $\lim_{x\to\infty}\frac{f(x)}{g(x)}=-\infty$ for $m>n$ and coef of $x^m<0$

iv) $\lim_{x\to\infty}\frac{f(x)}{g(x)}=\frac{\text{coef of } x^m \text{ in Nr}}{\text{coef of } x^n \text{ in Dr}}$ for $m=n$

➢ $\lim_{x\to 0^+} e^{\frac{1}{x}}=\infty, \lim_{x\to 0^-} e^{\frac{1}{x}}=0$

➢ $\lim_{x\to 0^+} e^{-\frac{1}{x}}=0, \lim_{x\to 0^-} e^{-\frac{1}{x}}=\infty$

➢ $\lim_{n\to\infty} x^n=0$ if $|x|<1$

➢ $\lim_{n\to\infty} x^n=\infty$ if $|x|>1$

W.E-1 : $\lim_{n\to\infty}\left(4^n+5^n\right)^{1/n}$ ***is equal to***

Sol: Given limit =

$$\lim_{n\to\infty}\left(4^n+5^n\right)^{1/n}=\lim_{n\to\infty}5\left(1+\left(\frac{4}{5}\right)^n\right)^{1/n}=5$$

$$\left(\because\left(\frac{4}{5}\right)^n\to 0 \text{ as } n\to\infty\right)$$

W.E-2 : $\lim_{n\to\infty}\frac{5^{n+1}+3^n-2^{2n}}{5^n+2^n+3^{2n+3}}$ ***is equal to***

Sol: $\lim_{n\to\infty}\frac{5^{n+1}+3^n-2^{2n}}{5^n+2^n+3^{2n+3}}=\lim_{n\to\infty}\frac{5.5^n+3^n-4^n}{5^n+2^n+27.9^n}$

$$=\lim_{n\to\infty}\frac{5.\frac{5^n}{9^n}+\frac{3^n}{9^n}-\frac{4^n}{9^n}}{\frac{5^n}{9^n}+\frac{2^n}{9^n}+27}=\frac{0+0-0}{0+0+27}=0$$

➢ $\lim_{x\to 0}\sin\left(\frac{1}{x}\right)=\lim_{x\to 0}\cos\left(\frac{1}{x}\right)=$ Does not exist

➢ $\lim_{x\to 0}x\sin\left(\frac{1}{x}\right)=\lim_{x\to 0}x\cos\left(\frac{1}{x}\right)=0$

L'Hospital's Rule:

➢ If $\operatorname{Lim}_{x\to a}\frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ then

$$\operatorname{Lim}_{x\to a}\frac{f(x)}{g(x)}=\operatorname{Lim}_{x\to a}\frac{f^1(x)}{g^1(x)}.$$

If $\operatorname{Lim}_{x\to a}\frac{f^1(x)}{g^1(x)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ then

$$\operatorname{Lim}_{x\to a}\frac{f(x)}{g(x)}=\operatorname{Lim}_{x\to a}\frac{f^{11}(x)}{g^{11}(x)}$$

If $\operatorname{Lim}_{x\to a}\frac{f^{11}(x)}{g^{11}(x)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ then

This can be continued till we finally arrive at a determinate result.

W.E-3 : Let $f(x)$ be a twice differentiable function and $f''(0)=5$, then $\lim_{x\to 0}\frac{3f(x)-4f(3x)+f(9x)}{x^2}$ is equal to

Sol: $\lim_{x\to 0}\frac{3f(x)-4f(3x)+f(9x)}{x^2}\left(\frac{0}{0}\text{ form}\right)$

$$=\lim_{x\to 0}\frac{3f'(x)-12f'(3x)+9f'(9x)}{2x}\left(\frac{0}{0}\text{ form}\right)$$

$$=\lim_{x\to 0}\frac{3f''(x)-36f''(3x)+81f''(9x)}{2}$$

$$=\frac{3f''(0)-36f''(0)+81f''(0)}{2}$$

$$=24f''(0)=24(5)=120$$

Some useful results:

➢ Let $S=\{x,\sin x,\tan x,\sinh x,\tanh x,\sin^{-1}x,$
$\tan^{-1}x,\sinh^{-1}x,\tanh^{-1}x\}$

If $f(x), g(x) \in S$ then $\underset{x\to 0}{Lt}\dfrac{f(mx)}{g(nx)}=\dfrac{m}{n}$, .

If $f_1(x), f_2(x), g_1(x), g_2(x) \in S$ then

$$\underset{x\to 0}{Lt}\frac{f_1(mx)\pm f_2(nx)}{g_1(px)\pm g_2(qx)}=\frac{m\pm n}{p\pm q}$$

➢ $\underset{x\to o}{Lt}\dfrac{\sin ax}{\tan bx}=\dfrac{a}{b}$, $\underset{x\to 0}{Lt}\dfrac{1-\cos ax}{x^2}=\dfrac{a^2}{2}$

W.E-4 : $\underset{x\to 0}{Lt}\dfrac{\sin 7x+\sin 5x}{\tan 5x-\tan 2x}=\dfrac{7+5}{5-2}=4$

➢ If $f_1(x), f_2(x), g_1(x), g_2(x) \in S$ and m + n = p + q then

$$\underset{x\to 0}{Lt}\frac{f_1^m(ax)f_2^n(bx)}{g_1^p(cx)g_2^q(dx)}=\frac{a^m b^n}{c^p d^q}$$

W.E-5 : $\underset{x\to 0}{Lt}\dfrac{\sin^3 2x\tan^2 3x}{x\sin^4 4x}=\dfrac{2^3\times 3^2}{4^4}=\dfrac{9}{32}$

➢ If $g_1(x), g_2(x) \in S$ then

$$\underset{x\to 0}{Lt}\frac{1-\cos ax}{g_1(cx)g_2(dx)}=\frac{a^2}{2cd}$$

W.E-6 : $\underset{x\to 0}{Lt}\dfrac{1-\cos x}{x\sin 3x}=\dfrac{1}{6}$

➢ If $g_1(x), g_2(x) \in S$ then

$$\underset{x\to 0}{Lt}\frac{1-\cos^n(ax)}{g_1(cx)g_2(dx)}=\frac{na^2}{2cd}$$

W.E-7 : $\underset{x\to 0}{Lt}\dfrac{1-\cos^3 2x}{\sin 5x\tan 7x}=\dfrac{3\times 2^2}{2\times 5\times 7}=\dfrac{6}{35}$

➢ If $g_1(x), g_2(x), ..., g_{2n}(x) \in S$ then

$$\underset{x\to 0}{Lt}\frac{1-\cos(ax^n)}{g_1(c_1x).g_2(c_2x)........g_{2n}(c_{2n}x)}=\frac{a^2}{2c_1c_2........c_{2n}}$$

W.E-8: $\underset{x\to 0}{Lt}\dfrac{1-\cos(2x^3)}{x\sin^2(2x)\tan^3(3x)}=\dfrac{4}{2\times 2^2\times 3^3}=\dfrac{1}{54}$

➢ If $g_1(x), g_2(x) \in S$ then

$$\underset{x\to 0}{Lt}\frac{\cos ax-\cos bx}{f(cx)g(dx)}=\frac{b^2-a^2}{2cd}$$

W.E-9 : $\underset{x\to 0}{Lt}\dfrac{\cos 3x-\cos 5x}{x^2}=\dfrac{25-9}{2}=8$

➢ If $g_1(x), g_2(x) \in S$ then

$$\underset{x\to 0}{Lt}\frac{\cos^n ax-\cos^n bx}{g_1(cx).g_2(dx)}=\frac{n(b^2-a^2)}{2cd}$$

W.E-10 : $\underset{x\to 0}{Lt}\dfrac{\cos^3 3x-\cos^3 5x}{x^2}=\dfrac{3(25-9)}{2}=24$

➢ If $g_1(x), g_2(x), ..., g_{2n}(x) \in S$ then

$$\underset{x\to 0}{Lt}\frac{\cos(ax^n)-\cos(bx^n)}{g_1(c_1x).g_2(c_2x).......g_{2n}(c_{2n}x)}=\frac{b^2-a^2}{2c_1c_2........c_{2n}}$$

W.E-11:

$$\underset{x\to 0}{Lt}\frac{\cos(2x^3)-\cos(5x^3)}{x\sin^2(2x)\tan^3(3x)}=\frac{25-4}{2\times 2^2\times 3^3}=\frac{7}{18\times 4}=\frac{7}{72}$$

➢ If $g_1(x) \in S$ then

$$\underset{x\to 0}{Lt}\frac{\tan^n(ax)-\sin^n(ax)}{[g(x)]^{n+2}}=\frac{na^{n+2}}{2}$$

W.E-12: $\underset{x\to 0}{Lt}\dfrac{\tan x-\sin x}{x^3}=\dfrac{1}{2}$

➢ $\underset{x\to 0}{Lt}\dfrac{\sqrt{1+x^n}-\sqrt{1-x^n}}{x^n}=1$

W.E-13: $\underset{x\to 0}{Lt}\dfrac{\sqrt{1+x^2}-\sqrt{1-x^2}}{x^2}=1$

➢ $\underset{x\to 0}{Lt}\dfrac{\sqrt[n]{a+x^m}-\sqrt[n]{a-x^m}}{x^m}=\dfrac{2}{n}a^{\frac{1}{n}-1}$

W.E-14 : $\underset{x\to 0}{Lt}\dfrac{\sqrt{a+x}-\sqrt{a-x}}{x}=a^{\frac{1}{2}-1}=\dfrac{1}{\sqrt{a}}$

➢ If $g(x) \in S$ then $\underset{x\to 0}{Lt}\dfrac{\sqrt{a+x}-\sqrt{a-x}}{g(x)}=\dfrac{1}{\sqrt{a}}$

W.E-15 : $\underset{x\to 0}{Lt}\dfrac{\sqrt{3+x}-\sqrt{3-x}}{\sin x}=\dfrac{1}{\sqrt{3}}$

➢ If $g(x) \in S$ then $\underset{x\to 0}{Lt}\dfrac{\sqrt{a+x}-\sqrt{a}}{g(x)}=\dfrac{1}{2\sqrt{a}}$

W.E-16 : $\underset{x\to 0}{Lt}\dfrac{\sqrt{2+x}-\sqrt{2}}{x}=\dfrac{1}{2\sqrt{2}}$

➢ $\underset{x\to 0}{Lt}\dfrac{x.a^{\alpha x}-x}{1-\cos(mx)}=\dfrac{2\alpha}{m^2}\log a$

W.E-17: $\underset{x\to 0}{Lt}\dfrac{x.2^{3x}-x}{1-\cos(3x)}=\dfrac{2\times 3}{3^2}\log 2=\dfrac{2}{3}\log 2$

➢ $\lim\limits_{x\to a}f(x)=1$ *and* $\lim\limits_{x\to a}g(x)=\infty$ then

$\lim\limits_{x\to a}[f(x)]^{g(x)}=e^{\lim\limits_{x\to a}g(x)[f(x)-1]}$

W.E-18 : If $\lim\limits_{x\to 0}\left(1+ax+bx^2\right)^{2/x}=e^3$, then the values of a and b are

Sol: Let $\lim\limits_{x\to 0}\left(1+ax+bx^2\right)^{2/x}$ is of the form 1^∞

$e^{\lim\limits_{x\to 0}\left(1+ax+bx^2-1\right).\frac{2}{x}}=e^{\lim\limits_{x\to 0}(2a+2bx)}=e^{2a}=e^3\,(given)$

$\therefore a=3/2$ *and* $b\in R$

➢ $\lim\limits_{x\to a}f(x)=0$ and $\lim\limits_{x\to a}g(x)=0$, then

$\lim\limits_{x\to a}[f(x)]^{g(x)}=e^{\lim\limits_{x\to a}g(x)\log f(x)}$ $\quad(f(x)>0)$

W.E-19: $\underset{x\to 1}{Lt}\left(1-x^2\right)^{\frac{1}{\log(1-x)}}=$

Sol : $e^{\lim\limits_{x\to 1}\frac{\log\left(1-x^2\right)}{\log(1-x)}}$

$=e^{\lim\limits_{x\to 1}\left[\frac{\log(1-x)+\log(1+x)}{\log(1-x)}\right]}$

$=e^{\lim\limits_{x\to 1}\left[1+\frac{\log(1+x)}{\log(1-x)}\right]}$

$=e^{1+0}$

$=e$

➢ $\lim\limits_{x\to 0}\left[\dfrac{a_1^x+a_2^x+a_3^x+...+a_n^x}{n}\right]^{\frac{1}{x}}=\left(a_1.a_2...a_n\right)^{\frac{1}{n}}$

➢ $\lim\limits_{x\to\infty}\left[\dfrac{a_1^{\frac{1}{x}}+a_2^{\frac{1}{x}}+a_3^{\frac{1}{x}}+...+a_n^{\frac{1}{x}}}{n}\right]^{x}=\left(a_1.a_2...a_n\right)^{\frac{1}{n}}$

W.E-20 : Evaluate $\lim\limits_{x\to 0}\left(\dfrac{2^x+2^{2x}+2^{3x}}{3}\right)^{1/x}$

Sol : $\lim\limits_{x\to 0}\left(\dfrac{2^x+2^{2x}+2^{3x}}{3}\right)^{1/x}$

$=(2.4.8)^{1/3}=4$

W.E-21 : $\lim\limits_{x\to 0}[\cos x+m\sin ax]^{\frac{n}{x}}=e^{amn}$

➢ $\lim\limits_{x\to\infty}a^x=\begin{cases}0, & 0\le a<1\\ 1, & a=1\\ \infty, & a>1\\ does\ not\ exists, & a<0\end{cases}$

Some frequently used expansions :

➢ i) $(1+x)^p=$

$1+px+\dfrac{p(p-1)}{2!}x^2+\dfrac{p(p-1)(p-2)}{3!}x^3+...\infty$, if $|x|<1$

ii) $e^x=1+\dfrac{x}{1!}+\dfrac{x^2}{2!}+\dfrac{x^3}{3!}+...\infty$

iii) $a^x=1+\dfrac{x}{1!}.\log_e a+\dfrac{x^2}{2!}(\log_e a)^2+...\infty$

iv) $\log_e(1+x)=x-\dfrac{x^2}{2}+\dfrac{x^3}{3}-\dfrac{x^4}{4}+...\infty$

v) $\log_e(1-x)=-\left(x+\dfrac{x^2}{2}+\dfrac{x^3}{3}+\dfrac{x^4}{4}+...\infty\right)$

vi) $\sin x=x-\dfrac{x^3}{3!}+\dfrac{x^5}{5!}-....\infty$

vii) $\cos x=1-\dfrac{x^2}{2!}+\dfrac{x^4}{4!}-.....\infty$

viii) $\tan x=x+\dfrac{x^3}{3}+\dfrac{2x^5}{15}.....\infty$

ix) $\sin^{-1}x = x+\frac{1}{2}.\frac{x^3}{3}+\frac{1}{2}.\frac{3}{4}.\frac{x^5}{5}+.....\infty$

x) $\tan^{-1}x = x-\frac{x^3}{3}+\frac{x^5}{5}-.....\infty$

W.E-22: Evaluate $\underset{x\to0}{Lim}\left(\frac{(1+x)^{\frac{1}{x}}-e+\frac{ex}{2}}{\sin^2 x}\right)$.

Sol : $\underset{x\to0}{Lt}\frac{(1+x)^{1/x}-e+\frac{ex}{2}}{\sin^2 x}$

$=\underset{x\to0}{Lt}\frac{(1+x)^{1/x}-e+\frac{ex}{2}}{x^2}.\left(\frac{x^2}{\sin^2 x}\right)$

$=\underset{x\to0}{Lt}\frac{(1+x)^{1/x}-e+\frac{ex}{2}}{\sin^2 x}.1$

$=\underset{x\to0}{Lt}\frac{e\left[1-\frac{x}{2}+\frac{11}{24}x^2....\right]-e+\frac{ex}{2}}{x^2}$

$=\frac{11e}{24}$.

W.E-23: **Find** $\underset{x\to0}{Lt}\frac{\left(\sin x-x+\frac{x^3}{6}\right)}{x^5}$.

Sol : $\underset{x\to0}{Lt}\frac{\left(\sin x-x+\frac{x^3}{6}\right)}{x^5}$

$=\underset{x\to0}{Lt}\frac{\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+.....\right)-x+\frac{x^3}{6}}{x^5}$

$=\underset{x\to0}{Lt}\frac{\left(\frac{x^5}{5!}-\frac{x^7}{7!}+.....\right)}{x^5}$

$=\underset{x\to0}{Lt}\left(\frac{1}{5!}-\frac{x^2}{7!}+\right.$ terms containing positive integral powers of x)

$=\frac{1}{5!}=\frac{1}{120}$.

W.E-24: The value of $\lim_{n\to\infty}\left[\sqrt[3]{(n+1)^2}-\sqrt[3]{(n-1)^2}\right]$ ***is***

Sol: $\lim_{n\to\infty}\left[\sqrt[3]{(n+1)^2}-\sqrt[3]{(n-1)^2}\right]$

$=\lim_{n\to\infty}n^{2/3}\left[\left(1+\frac{1}{n}\right)^{2/3}-\left(1-\frac{1}{n}\right)^{2/3}\right]$

$=\lim_{n\to\infty}n^{2/3}\left[\left(1+\frac{2}{3}.\frac{1}{n}+\frac{\frac{2}{3}\left(\frac{2}{2}-1\right)}{2!}\frac{1}{n^2}....\right)\right.$

$\left.-\left(1-\frac{2}{3}.\frac{1}{n}+\frac{\frac{2}{3}\left(\frac{2}{3}-1\right)}{2!}\frac{1}{n^2}....\right)\right]$

$=\lim_{n\to\infty}n^{2/3}\left[\frac{4}{3}.\frac{1}{n}+\frac{8}{81}.\frac{1}{n^3}+....\right]$

$=\underset{n\to\infty}{Lim}\left[\frac{4}{3}.\frac{1}{n^{1/3}}+\frac{8}{81}.\frac{1}{n^{7/3}}+....\right]=0$

Sandwich theorem or Squeeze principle:

➢ If f, g, h are functions such that

$f(x)\le g(x)\le h(x)$

then $\underset{x\to a}{Lt}f(x)\le\underset{x\to a}{Lt}g(x)\le\underset{x\to a}{Lt}h(x)$ and

$\underset{x\to a}{Lt}f(x)=\underset{x\to a}{Lt}h(x)=l$ then $\underset{x\to a}{Lt}g(x)=l$

W.E-25: ***If*** $3-\frac{x^2}{12}\le f(x)\le 3+\frac{x^3}{9}$ ***for all*** $x\ne 0$***,***

then the value of $\lim_{x\to 0} f(x)$ ***is***

Sol: According to the equation,

$$\lim_{x\to 0}\left(3-\frac{x^2}{12}\right)\le \lim_{x\to 0} f(x)\le \lim_{x\to 0}\left(3+\frac{x^3}{9}\right)$$

$$\Rightarrow (3-0)\le \lim_{x\to 0} f(x)\le (3+0)$$

Hence, $\lim_{x\to 0} f(x)=3$

➢ $$\underset{n\to\infty}{Lt}\frac{[1^k x]+[2^k x]+.....[n^k x]}{n^{k+1}}=\frac{x}{k+1}(k\in N)$$

W.E-26: **Show that** $\underset{n\to\infty}{Lt}\frac{[x]+[2x]+....[nx]}{n^2}=\frac{x}{2}$.

Sol : For r=1,2,3.....n, $r.x-1<[rx]\le rx$

$$\Rightarrow \sum_{r=1}^{n}(rx-1)<\sum_{r=1}^{n}[rx]\le\sum_{r=1}^{n}(rx)$$

$$\Rightarrow \frac{\frac{n(n+1)x}{2}-n}{n^2}<\frac{\sum_{r=1}^{n}[rx]}{n^2}\le\frac{n(n+1)x}{2n^2}$$

$$\Rightarrow \underset{n\to\infty}{Lt}\left(\left(1+\frac{1}{n}\right)\frac{x}{2}-\frac{1}{n}\right)\le \underset{n\to\infty}{Lt}\frac{\sum_{r=1}^{n}[rx]}{n^2}\le \underset{n\to\infty}{Lt}\left(1+\frac{1}{n}\right)\frac{x}{2}$$

(Note that x is a constant and n is a variable)

$$\Rightarrow \frac{x}{2}\le \underset{n\to\infty}{Lt}\frac{[1.x]+[2.3]+.....+[n.x]}{n^2}\le\frac{x}{2}$$

∴ By sandwich theorem,

$$\underset{n\to\infty}{Lt}\frac{[1x]+[2x]+....[nx]}{n^2}=\frac{x}{2}.$$

LEVEL - I (C.W)

Evaluation of algebra of limits :

1. $\underset{x\to 0}{Lim}\frac{\sqrt[K]{1+x}-1}{x}$ (K **is a positive integer)**

1) K 2) $-K$ 3) $\frac{1}{K}$ 4) $-\frac{1}{K}$

2. $\underset{x\to 1}{Lim}\frac{(2x-3)(\sqrt{x}-1)}{2x^2+x-3}=$

1) $\frac{1}{10}$ 2) $-\frac{1}{10}$ 3) $\frac{2}{5}$ 4) $-\frac{2}{5}$

3. $\underset{x\to 0}{Lim}\frac{\sqrt[3]{1+\sin x}-\sqrt[3]{1-\sin x}}{x}=$

1) 0 2) 1 3) $\frac{2}{3}$ 4) $\frac{3}{2}$

4. **If** $\underset{x\to 5}{Lim}\frac{x^k-5^k}{x-5}=500$**, then the positive integral value of k is**

1) 3 2) 4 3) 5 4) 6

5. $\underset{x\to 1}{Lim}\frac{\sqrt{x^2-1}+\sqrt{x-1}}{\sqrt{x^2-1}}=$

1) $1+\frac{1}{\sqrt{2}}$ 2) $1-\frac{1}{\sqrt{2}}$

3) $-1+\frac{1}{\sqrt{2}}$ 4) $-1-\frac{1}{\sqrt{2}}$

6. **If a > 0 and** $\underset{x\to a}{Lim}\frac{a^x-x^a}{x^x-a^a}=-1$ **then a =**

1) 0 2) 1 3) e 4) 2e

7. $\underset{x\to 0}{Lim}\frac{x\left(1-\sqrt{1-x^2}\right)}{\sqrt{1-x^2}\left(\sin^{-1}(x)\right)^3}=$

1) 1 2) $\frac{1}{2}$ 3) $-\frac{1}{2}$ 4) -1

8. **If** $\lim_{x\to 0}\left(\frac{\cos 4x+a\cos 2x+b}{x^4}\right)$ **is finite then the value of** a,b **respectively**

1) 5 2) -5,-4 3) -4,3 4) 4,5

9. $\lim_{x\to 1}\left|\sec\left(\frac{\pi x}{2}\right)\log x\right|$ **is**

1) $-\frac{2}{\pi}$ 2) $-\frac{\pi}{2}$ 3) $\frac{2}{\pi}$ 4) $\frac{\pi}{2}$

10. $\lim_{x\to 0}\frac{\sqrt{1+x^2}-\sqrt{1-x+x^2}}{3^x-1}=$ (Eamcet2014)

1) log9 2) $\frac{1}{\log 9}$ 3) $\log 3$ 4) $\frac{1}{\log 3}$

Evaluation of left & right hand limits :

11. $\underset{x\to 0}{Lim}\frac{\sin x}{\sqrt{x^2}}=$

1) 1 2) -1
3) 0 4) doesn't exist

12. $\lim_{x\to 2^+}\left(\frac{[x]^3}{3}-\left[\frac{x}{3}\right]^3\right)$ **is (where [ ] is g.i.f)**

1) 0 2) $\frac{64}{27}$ 3) $\frac{8}{3}$ 4) $\frac{10}{3}$

13. **If** $f:R\to R$ **is defined by (where [ ] is g.i.f)** $f(x)=[x-3]+|x-4|$ **for** $x\in R$ **then** $\lim_{x\to 3^-}f(x)=$ **[EAM- 2008]**

1) -2 2) -1 3) 0 4) 2.

Evaluation of trigonometric limits :

14. $\lim_{x\to 0}\frac{e^x-e^{\sin x}}{2(x-\sin x)}=$ **[EAMCET 2007]**

1) -1/2 2) 1/2 3) 1 4) 3/2

15. $\underset{x\to 0}{Lim}\frac{1-\cos^3 x}{x\sin 2x}=$

1) 1/2 2) 3/2 3) 3/4 4)1/4

16. $\underset{x\to 1}{Lim}(1-x)Tan\left(\frac{\pi x}{2}\right)=$

1) π 2) 2π 3) $\frac{\pi}{2}$ 4) $\frac{2}{\pi}$

17. $\underset{x\to\frac{\pi}{6}}{Lim}\frac{3\sin x-\sqrt{3}\cos x}{6x-\pi}=$

1) $\sqrt{3}$ 2) $\frac{1}{\sqrt{3}}$ 3) $-\sqrt{3}$ 4) $-\frac{1}{\sqrt{3}}$

18. $\underset{x\to\frac{\pi}{2}}{Lim}\frac{\cot x-\cos x}{\left(\frac{\pi}{2}-x\right)^3}=$

1) $\frac{-1}{2}$ 2) $\frac{1}{2}$ 3) 2 4) -2

19. $\underset{x\to 0}{Lim}\frac{\sec 4x-\sec 2x}{\sec 3x-\sec x}=$

1) $\frac{3}{2}$ 2) $\frac{2}{3}$ 3) $\frac{1}{3}$ 4) $\frac{3}{4}$

20. $\underset{x\to 0}{Lim}\frac{3\sin(x^g)-\sin(3x^g)}{x^3}=$

1) $\left(\frac{\pi}{200}\right)^3$ 2) $4\left(\frac{\pi}{200}\right)^3$

3) $\frac{\pi}{200}$ 4) $\frac{\pi}{100}$

21. $\lim_{x\to 0}\frac{\sin(\pi\cos^2 x)}{x^2}=$ **(MAINS-2014)**

1) $-\pi$ 2) π 3) $\frac{\pi}{2}$ 4) 1

Evaluation of exponential & logarithemic limits :

22. $\underset{x\to 0}{Lim}\log\left|\frac{\log(1+x)}{x}\right|=$

1) 0 2) 1 3) e 4) 1/e

23. **The value of** $\lim_{x\to 2}\frac{2^x+2^{3-x}-6}{\sqrt{2^{-x}-2^{1-x}}}$ **is**

1) 16 2) 8 3) 4 4) 2

24. $\underset{x\to 0}{Lim}\frac{e^x+\sin x-1}{\log(1+x)}=$

1) 1 2) $\frac{1}{3}$ 3) $\frac{2}{3}$ 4) 2

25. $\lim_{x\to 0}\frac{10^x-2^x-5^x+1}{x\tan x}$ **is**

1) $\log 2$ 2) $\frac{\log 2}{\log 5}$

3) $(\log 2)(\log 5)$ 4) $\log 10$

Limits of the form tends to infinite :

26. $\underset{n\to\infty}{Lim}\left(\frac{1}{a(a+d)}+\frac{1}{(a+d)(a+2d)}+\frac{1}{(a+2d)(a+3d)}+.....to\ n\ terms\right)$

$(a>0,d>0)$ =

1) $\frac{1}{a}$ 2) $\frac{1}{d}$ 3) $\frac{a}{d}$ 4) $\frac{1}{ad}$

27. $\underset{n\to\infty}{Lim}\frac{1+\frac{1}{2}+\frac{1}{4}+....+\frac{1}{2^n}}{1+\frac{1}{3}+\frac{1}{9}+....+\frac{1}{3^n}}=$

1) 4/3 2) 3/4 3) 1/2 4) 0

28. $\underset{x\to\infty}{Lim}\left(\frac{x^2\sin\left(\frac{1}{x}\right)-x}{1-|x|}\right)=$

1) 0 2) 1 3) -1 4) 2

29. **If** $o<h<q$ **then** $\lim_{n\to\infty}\left(q^n+h^n\right)^{\frac{1}{n}}=$

1) e 2) h 3) q 4) 0

30. $\underset{x\to\infty}{Lim}\left(\sin\sqrt{x+1}-\sin\sqrt{x}\right)=$

1) 2 2) -2 3) 0 4) 1

31. $\underset{x\to\infty}{Lim}\, x\cos\left(\frac{\pi}{8x}\right)\sin\left(\frac{\pi}{8x}\right)=$

1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{8}$ 4) $\frac{\pi}{4}$

32. $\underset{x\to\infty}{Lim}\left(\sqrt{x^2+ax+a^2}-\sqrt{x^2+a^2}\right)=$

1) 0 2) $\frac{a}{2}$ 3) $-\frac{a}{2}$ 4) a

33. $\underset{x\to\infty}{Lim}\frac{x-\log x}{x+\log x}=$

1) 1 2) -1 3) 0 4) 2

34. **For a>1 then** $\underset{x\to\infty}{Lim}\frac{a^x-a^{-x}}{a^x+a^{-x}}=$

1) 1 2) 1/2 3) 1/3 4) 1/15

35. $\underset{x\to\infty}{Lim}\frac{2x+7\sin x}{4x+3\cos x}=$

1) 1 2) -1 3) 1/2 4) -1/2

36. $\underset{x\to\infty}{Lim}\frac{8|x|+3x}{3|x|-2x}=$

1) 11 2) 8 3) 0 4) $\frac{1}{8}$

37. $\underset{x\to\infty}{Lim}\frac{(x+1)^{10}+(x+2)^{10}+.....+(x+100)^{10}}{x^{10}+10^{10}}=$

1) 10 2) 100 3) 1000 4) 1

38. $\underset{n\to\infty}{Lim}\frac{3.2^{n+1}-4.5^{n+1}}{5.2^n+7.5^n}=$

1) $-20/7$ 2) 20/7 3) 10/7 4) $-10/7$

39. $\underset{n\to\infty}{Lim}\frac{1}{n^3}\left\{1+3+6+..........+\frac{n(n+1)}{2}\right\}=$

1) 0 2) 2 3) $\frac{1}{6}$ 4) $\frac{1}{3}$

Limit of the form $0^0,1^\infty,\infty^0$:

40. $\underset{x\to 1}{Lim}(2-x)^{\tan\left(\frac{\pi x}{2}\right)}=$

1) $e^{\frac{1}{\pi}}$ 2) $e^{\frac{2}{\pi}}$ 3) $-e^{\frac{2}{\pi}}$ 4) e

41. $\underset{x\to 0+}{Lt}(Sinx)^{\tan x}=$

1) e 2) e^2 3) -1 4) 1

42. $\underset{n\to\infty}{Lim}(\pi n)^{2/n}=$

1) 0 2) 1 3) 2 4) 3

43. $\lim\limits_{x\to\infty}\left(\dfrac{a^{1/x}+b^{1/x}+c^{1/x}}{3}\right)^x$ = **(where a,b,c are real and non-zero) [EAMCET 1998]**

1) 0 2) $(abc)^{1/3}$ 3) $(abc)^{-1/3}$ 4) 1

44. $\lim\limits_{n\to\infty}\left(1+\sin\left(\dfrac{a}{n}\right)\right)^n =$

1) e 2) e^a 3) a^e 4) a

L.H.rule :

45. **If** $f(9)=9,\ f^1(9)=4,\ \lim\limits_{x\to 9}\dfrac{\sqrt{f(x)}-3}{\sqrt{x}-3} =$

1) 4 2) $\dfrac{1}{4}$ 3) $\dfrac{1}{2}$ 4) $-\dfrac{1}{2}$

46. **If** $f(a)=2, f^1(a)=1, g(a)=-1, g^1(a)=2,$ **then** $\lim\limits_{x\to a}\dfrac{g(x)f(a)-g(a)f(x)}{x-a} =$

1) $\dfrac{1}{5}$ 2) 5 3) $-\dfrac{1}{5}$ 4) -5

47. **If** $f(x)=xTan^{-1}x$ **then** $\underset{x\to 1}{Lt}\dfrac{f(x)-f(1)}{x-1} =$ (Eamcet-2014)

1) $\dfrac{\pi}{4}$ 2) $\dfrac{\pi+1}{4}$ 3) $\dfrac{\pi+2}{4}$ 4) $\dfrac{\pi+3}{4}$

Problems using expansions:

48. $\underset{x\to 0}{Lt}\dfrac{1-\cos x\cos 2x\cos 3x}{\sin^2 2x} =$

1) $\dfrac{3}{2}$ 2) $\dfrac{5}{2}$ 3) $\dfrac{7}{4}$ 4) $\dfrac{9}{2}$

Problems using sandwich theorem:

49. $\lim\limits_{x\to 0} x^3\cos\dfrac{2}{x} =$

1) 0 2) 1
3) ∞ 4) does not exist

50. $\lim\limits_{n\to\infty}\left(\dfrac{n}{n^2+1}+\dfrac{n}{n^2+2}+\dfrac{n}{n^2+3}+\ldots+\dfrac{n}{n^2+n}\right) =$

1) 0 2) 1
3) ∞ 4) does not exist

LEVEL-I (C.W)-KEY

1) 3	2) 2	3) 3	4) 2
5) 1	6) 2	7) 2	8) 3
9) 1	10) 2	11) 4	12) 3
13) 3	14) 2	15) 3	16) 4
17) 2	18) 2	19) 1	20) 2
21 2	22) 1	23) 2	24) 4
25) 3	26) 4	27) 1	28) 1
29) 3	30) 3	31) 3	32) 2
33) 1	34) 1	35) 3	36) 1
37) 2	38) 1	39) 3	40) 2
41) 4	42) 2	43) 2	44) 2
45) 1	46) 2	47) 3	48)3
49) 1	50) 2		

LEVEL-I (C.W)-HINTS

1. Using L-Hospital rule
2. Given limit

$$\lim_{x\to 0}\frac{(2x-3)(\sqrt{x}-1)}{(\sqrt{x}-1)(\sqrt{x}+1)(2x+3)}=\frac{(-1)}{2(5)}=\frac{-1}{10}$$

3. Using L-Hospital rule
4. Using $\lim\limits_{x\to a}\dfrac{x^p-a^p}{x^q-a^q}=\dfrac{p}{q}a^{p-q}$
5. $\lim\limits_{x\to 1}\dfrac{\sqrt{x^2-1}+\sqrt{x-1}}{\sqrt{x^2-1}}=\lim\limits_{x\to 1}\dfrac{\sqrt{(x-1)[x+1]}+\sqrt{x-1}}{\sqrt{(x-1)(x+1)}}$

$$=\frac{\sqrt{2}+1}{\sqrt{2}}$$

6. Use L-Hospital rule

$$\frac{d}{dx}(a^x)=a^x\log a,\ \frac{d}{dx}(x^a)=a.x^{a-1}$$

$$\frac{d}{dx}(x^x)=x^x(1+\log x)$$

7. On rationalising given limit

$$=\lim_{x\to 0}\frac{x^3}{[\sin^{-1}(x^3)]}.\frac{1}{1+\sqrt{1-x^2}}$$

8. By L-Hospital rule

9. $\lim_{x\to1}\left[\sec\left(\frac{\pi x}{2}\right)\log x\right]$

$=\lim_{x\to1}\frac{\log x}{\cos\left(\frac{\pi x}{2}\right)}=\left(\frac{0}{0}\right)$

$=\lim_{x\to1}\frac{\frac{1}{x}}{-\sin\left(\frac{\pi x}{2}\right).\frac{\pi}{2}}=-\frac{1}{\frac{\pi}{2}\sin\frac{\pi}{2}}=-\frac{2}{\pi}$

10. On rationilising

11. $\lim_{x\to0}\frac{\sin x}{|x|}=$ does not exist

12. For

$x\in(2,3),[x]=2,\frac{x}{3}\in\left(\frac{2}{3},1\right)\Rightarrow\left[\frac{x}{3}\right]=0$

$\therefore\lim_{x\to2^+}\left(\frac{[x]^3}{3}-\left[\frac{x}{3}\right]^3\right)=\frac{1}{3}(2)^3-(0)^3=\frac{8}{3}$

13. Definition of step function

14. $\lim_{x\to0}\frac{e^{\sin x}\left(e^{x-\sin x}-1\right)}{2\left(x-\sin x\right)}=\frac{1}{2}$

15. Given limit is

$\lim_{x\to0}\frac{\left(1-\cos x\right)\left(1+\cos x+\cos^2 x\right)}{x\sin 2x}=\frac{3}{4}$

16. $\lim_{x\to1}\frac{(1-x)}{\cot\left(\frac{\pi x}{2}\right)}=\frac{2}{\pi}\lim_{x\to1}\frac{-1}{-\operatorname{cosec}^2\left(\frac{\pi x}{2}\right)}=\frac{2}{\pi}$

17. By L-Hospital rule

18. $\lim_{\theta\to0}\frac{\frac{\sin\theta}{\cos\theta}-\sin\theta}{\theta^3}=\lim_{\theta\to0}\left(\frac{\frac{\sin\theta}{\cos\theta}}{\theta}\right)\left(\frac{1-\cos\theta}{\theta^2}\right)$

19. $\lim_{x\to0}\frac{\sec 4x-\sec 2x}{\sec 3x-\sec x}=\frac{4^2-2^2}{3^2-1^2}=\frac{3}{2}$

20. $\lim_{x\to0}\frac{4\sin^3\left(\frac{\pi}{200}x\right)}{x^3}=4\left(\frac{\pi}{200}\right)^3$

21. $\lim_{x\to0}\frac{\sin\left(\pi-\pi\sin^2 x\right)}{x^2}$

22. $\lim_{x\to0}(1+x)^{\frac{1}{x}}=e$

23. $\lim_{x\to2}\frac{2^x+2^{3-x}-6}{\sqrt{2^{-x}}-2^{1-x}}$

$=\lim_{x\to2}\frac{(2^x)^2-6.2^x+2^3}{\sqrt{2^x}-2}$

[Multiplying N^r and D^r by 2^x]

$=\lim_{x\to2}(2^x-2)(\sqrt{2^x}+2)=(2^2-2)(2+2)=8$

24. $\lim_{x\to0}\frac{e^x+\sin x-1}{\log(1+x)}=\lim_{x\to0}\frac{e^x+\cos x}{\frac{1}{1+x}}=2$

25. $\lim_{x\to0}\frac{\left(5^x-1\right)\left(2^x-1\right)}{x\tan x}$

26. $\lim_{n\to\infty}\frac{1}{d}\left[\frac{1}{a}-\frac{1}{a+d}+\frac{1}{a+d}-\frac{1}{a+2d}+....\right.$

$\left.\frac{1}{a+(n-1)d}-\frac{1}{a+nd}\right]$

27. Given limit (if r < 1 then sum of terms in G.P is $a\left(\frac{1-r^n}{1-r}\right)$

$=\lim_{n\to\infty}\left(\frac{1+\frac{1}{2}+\frac{1}{4}+...+\infty}{1+\frac{1}{3}+\frac{1}{9}+...+\infty}\right)=\frac{4}{3}$

28. Divide with x

29. Taking q^n common

30. Using the formula sinC- sinD and sinx is bounded

31. $\lim_{x\to\infty}\frac{x.2\sin\left(\frac{\pi}{8x}\right)\cos\left(\frac{\pi}{8x}\right)}{2}$

$= \lim_{x\to\infty} \frac{x.\sin\left(\frac{\pi}{4x}\right)}{2} = \frac{\left(\frac{\pi}{4}\right)}{2}$

32. Rationalising
33. Use L-Hospital rule
34. Divide with a^x
35. Divide with x
36. $\underset{x\to\infty}{Lim} \frac{8|x|+3x}{3|x|-2x} = \underset{x\to\infty}{Lim} \frac{8x+3x}{3x-2x} = 11$
37. Divide with x^{10}
38. Taking 5^n common and simplify
39. Given limit is $\Sigma \frac{n^2+n}{2n^3}$
40. $\lim_{x\to1}(2-x)^{\tan\left(\frac{\pi x}{2}\right)} = e^{\lim_{x\to1}\tan\left(\frac{\pi x}{2}\right)(2-x-1)} = e^{\lim_{x\to1}\frac{(1-x)}{\cot\left(\frac{\pi x}{2}\right)}}$
    and using L hospital rule.
41. $\lim_{x\to0}(\sin x)^{\tan x} = \lim_{x\to0} e^{\tan x\log(\sin x)} = e^0 = 1$
42. apply log on both sides
43. $(abc)^{1/3}$
44. $e^{\lim_{n\to\infty} n\left(1+\sin\left(\frac{a}{n}\right)-1\right)} = e^a$
45. $\lim_{x\to9} \frac{\frac{1}{2\sqrt{f(x)}}.f'(x)}{\frac{1}{2\sqrt{x}}} = \frac{f'(9)}{\sqrt{f(9)}}\times\sqrt{9} = \frac{4}{3}\times3 = 4$
46. $f(a)=2, f'(a)=1, g(a)=-1, g'(a)=2$

    $\lim_{x\to a}\frac{g(x)f(a)-g(a)f(x)}{x-a}$

    By $L-H$ $Rule$

    $=g'(a)f(a)-g(a)f'(a)$

    $=(2)(2)-(-1)(1)=5$
47. Find $f^1(x)$ use L.H.Rule
48. $\lim_{x\to0}\frac{1-\left(1-\frac{x^2}{2}\right)\left(1-\frac{4x^2}{2}\right)\left(1-\frac{9x^2}{2}\right)}{4x^2} = \frac{7}{4}$
49. $-1 \le \cos\frac{2}{x} \le 1 \Rightarrow -x^3 \le x^3\cos\frac{2}{x} \le x^3$ for $x>0$ and $x^3 \le x^3\cos\frac{2}{x} \le -x^3$ for $x<0$.
    where $\underset{x\to0}{Lim}\, x^3\cos\frac{2}{x} = 0$ (or) $0\times$ finite number between -1 and +1 = 0
50. Use Sandwich Theorem

LEVEL - I (H.W)

Evaluation of algebra of limits :

1. $\underset{x\to a}{Lim}\frac{x^{\frac{5}{8}}-a^{\frac{5}{8}}}{x^{\frac{1}{3}}-a^{\frac{1}{3}}} =$

1) $\frac{15}{8}a^{\frac{7}{24}}$ 2) $\frac{15}{4}a^{\frac{7}{24}}$ 3) $-\frac{15}{8}a^{\frac{7}{24}}$ 4) $\frac{15}{4}a^{-\frac{7}{24}}$

2. $\underset{x\to2}{Lim}\frac{7x^2-11x-6}{3x^2-x-10} =$

1) $\frac{17}{11}$ 2) $\frac{11}{17}$ 3) $\frac{17}{14}$ 4) $-\frac{17}{11}$

3. $\underset{x\to0}{Lim}\frac{(1+x)^{\frac{1}{m}}-(1-x)^{\frac{1}{m}}}{(1+x)^{\frac{1}{n}}-(1-x)^{\frac{1}{n}}} =$

1) 1 2) $\frac{m+n}{m-n}$ 3) $\frac{n}{m}$ 4) $\frac{m}{n}$

4. $\underset{x\to0}{Lim}\frac{\sqrt{4+x}-\sqrt[3]{8+3x}}{x} =$

1) $-\frac{1}{2}$ 2) $\frac{1}{2}$ 3) -3 4) 0

5. $\underset{x\to0}{Lim}\frac{\sqrt{1+Tanx}-\sqrt{1-Tanx}}{\sin x} =$

1) 0 2) 1 3) 2 4) $\frac{1}{2}$

6. $\underset{x\to -1}{Lim}\dfrac{x+1}{\sqrt{x^2+3}-2}=$

1) -2 2) 1/2 3) 2 4) 0

7. **If** $7-\dfrac{x^2}{12}\le f(x)\le 7+\dfrac{x^3}{5}$ **for all** $x\ne 0$,

then $\underset{x\to 0}{Lim}\, f(x)=$

1) 3 2) 5 3) 7 4) 9

Evaluation of left & right hand limits :

8. **If** $f(x)=\begin{cases}\dfrac{x^2}{2}, & if\ 0\le x<1\\ 2x^2-2x+\dfrac{3}{2}, & if\ 1\le x\le 2\end{cases}$

then $\lim\limits_{x\to 1} f(x)=$

1) 1/2 2) 3/2
3) does not exist 4) -1/2

9. $\lim\limits_{x\to 0}\dfrac{\sin[\cos x]}{1+[\cos x]}$ **is (where [ ] is g.i.f)**

1) 1 2) 0
3) does not exist 4) 2

Evaluation of trigonometric limits :

10. $\underset{x\to 0}{Lim}\left(\dfrac{\sec ax-\sec bx}{x^2}\right)=$

1) $\dfrac{a^2-b^2}{2}$ 2) $\dfrac{b^2-a^2}{2}$
3) 0 4) 1

11. $\underset{x\to\frac{\pi}{4}}{Lim}\dfrac{\cos x-\sin x}{\left(\dfrac{\pi}{4}-x\right)(\cos x+\sin x)}=$

1) 2 2) 1 3) 0 4) 3

12. $\underset{x\to 0}{Lim}\dfrac{\sin x\sin\left(\dfrac{\pi}{3}+x\right)\sin\left(\dfrac{\pi}{3}-x\right)}{x}=$

1) $\dfrac{3}{4}$ 2) $\dfrac{1}{4}$ 3) $\dfrac{4}{3}$ 4) 0

13. $\underset{x\to 0}{Lim}\dfrac{(1-\cos 2x)\sin 5x}{x^2\sin 3x}=$

1) 10/3 2) 3/10 3) 6/5 4) 56

14. $\underset{x\to 5}{Lim}\dfrac{\sin^2(x-5)\tan(x-5)}{(x^2-25)(x-5)}=$

1) 1 2) 1/10 3) 0 4)-6

15. $\underset{x\to 0}{Lim}\dfrac{1-\cos 2x^\circ}{x^2}=$

1) 0 2) 1 3) $\left(\dfrac{\pi}{180}\right)^2$ 4) $2.\left(\dfrac{\pi}{180}\right)^2$

16. $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x\log(1+x)}=$

1) 1 2) 0
3) -1 4) $\dfrac{1}{2}$

17. $\underset{x\to 0}{Lt}\dfrac{1}{x^8}\left[1-\cos\left(\dfrac{x^2}{2}\right)\right]\left[1-\cos\left(\dfrac{x^2}{4}\right)\right]$

1) $\dfrac{1}{8}$ 2) $\dfrac{1}{8^2}$ 3) $\dfrac{2}{8^3}$ 4) $\dfrac{2}{8^4}$

18. $\underset{x\to\frac{\pi}{4}}{Lt}\dfrac{1-\cot^3 x}{2-\cot x-\cot^3 x}=$

1) $\dfrac{11}{4}$ 2) $\dfrac{3}{4}$ 3) $\dfrac{1}{2}$ 4) 0

Evaluation of exponential & logarithemic limits :

19. $\underset{x\to 0}{Lim}\dfrac{\log_e(1+x)}{3^x-1}=$

1) $\log_e 3$ 2) 0 3) 1 4) $\log_3 e$

20. $\underset{x\to 0}{Lim}\dfrac{e^{\alpha x}-e^{\beta x}}{\sin\alpha x-\sin\beta x}=$

1) 0 2) 1/2 3) 1/3 4) 1

21. $\underset{x\to 0}{Lim}\dfrac{e^{2x}-1}{3x}=$

1) 2/3 2) 6 3) 3/2 4) 1/6

22. $\underset{x\to 0}{Lim}\dfrac{\log(1+ax)-\log(1+bx)}{x}=$

1) $a+b$ 2) $a-b$ 3) $-(a+b)$ 4) ab

23. $\lim_{x\to 0}\frac{4^x-9^x}{x(4^x+9^x)}=$

1) $\log\left(\frac{3}{2}\right)$ 2) $\log\left(\frac{2}{3}\right)$

3) $\log\left(\frac{4}{3}\right)$ 4) $\log 2$

24. $\lim_{x\to 0}\frac{a^x-1}{b^x-1}=$

1) $\log_b a$ 2) $\log_a b$

3) $\log_e ab$ 4) $\log_e\left(\frac{a}{b}\right)$

Limits of the form tends to infinite :

25. $\lim_{x\to\infty} 2x\left(\sqrt{x^2+1}-x\right)=$

1) 1 2) 1/2 3) 0 4) -1

26. $\lim_{x\to\infty}\frac{\sqrt[7]{x^7-1}+\sqrt[5]{x^5+2}+\sqrt[9]{x^9-2}}{\sqrt[6]{x^6+1}+\sqrt[5]{x^5+1}-\sqrt[4]{x^4+4}}=$

1) 3 2) – 3 3) 1/3 4) – 1/3

27. $\underset{x\to\infty}{Lt}\, 5^x\sin\left(\frac{a}{5^x}\right)=$

1) 1 2) a 3) ∞ 4) not defined

28. $\lim_{n\to\infty}\frac{2^{\frac{1}{n}}-1}{2^{\frac{1}{n}}+1}=$

1) 1 2) $\frac{1}{2}$ 3) -1 4) 0

29. $\lim_{n\to\infty}\frac{(1+2+3+......+n\,terms)(1^2+2^2+......n\,terms)}{n(1^3+2^3+.....+n\,terms)}=$

1) $\frac{3}{2}$ 2) $\frac{2}{3}$ 3) 1 4) 0

30. $\lim_{n\to\infty}\left(\frac{1}{1.3}+\frac{1}{3.5}+....+\frac{1}{(2n-1)(2n+1)}\right)=$

1) 1 2) $\frac{1}{2}$ 3) $\frac{1}{3}$ 4) $\frac{1}{4}$

31. $\lim_{n\to\infty}\frac{1+3+5+.....+(2n-1)}{2+4+6+....+2n}=$

1) 0 2) 1 3) -1 4) 5

32. $\lim_{n\to\infty}\frac{(n+2)!+(n+1)!}{(n+3)!}=$

1) 1 2) 1/2 3) 0 4) .2

Limit of the form $0^0, 1^\infty, \infty^0$:

33. **If** $\underset{x\to\infty}{Lt}\left(1+\frac{\lambda}{x}+\frac{\mu}{x^2}\right)^{2x}=e^2$ **then**

1) $\lambda=1,\mu=2$ 2) $\lambda=2,\mu=1$

3) $\lambda=1,\mu=$ any real constant

4) $\lambda=\mu=1$

34. $\lim_{x\to\infty}\left(\frac{x+a}{x+b}\right)^{x+b}$

1) 1 2) e^{b-a} 3) e^{a-b} 4) e^b

35. $\lim_{x\to\pi}(1-4\tan x)^{\cot x}=$

1) e 2) e^4 3) e^{-1} 4) e^{-4}

36. $\lim_{x\to\infty}\left(\frac{x^2+1}{x^2-1}\right)^{x^2}=$

1) e 2) 1/e 3) e^2 4) e^{-2}

37. $\lim_{x\to\infty}\left(\frac{3x-4}{3x+2}\right)^{\frac{x+1}{3}}=$

1) $e^{-2/3}$ 2) $e^{3/2}$ 3) $e^{2/3}$ 4) e

38. $\lim_{x\to 1} x^{\left(\frac{1}{1-x^2}\right)}=$

1) $e^{-\frac{1}{2}}$ 2) $e^{\frac{2}{3}}$ 3) $e^{-\frac{3}{2}}$ 4) e

39. $\lim_{x\to 1}\left(\frac{1+x}{2+x}\right)^{\left(\frac{1-\sqrt{x}}{1-x}\right)}=$

1) $\frac{2}{3}$ 2) $\sqrt{\frac{2}{3}}$ 3) $\frac{2}{\sqrt{3}}$ 4) $\frac{\sqrt{2}}{3}$

L.H.rule :

40. **If** $f(x)=x^2+x+1$, **then** $\lim_{x\to 1}\frac{f(x)-f(1)}{x-1}=$

1) 3 2) 0 3) -1 4) 2

41. **f(x) = x²**, $\phi(a)=b^2$, $f'(a)=n\phi'(a)$ **and** $\Phi'(a)\neq 0$ **then** $\underset{x\to a}{Lim}\left(\dfrac{\sqrt{f(x)}-a}{\sqrt{\phi(x)}-b}\right)=$

1) $\dfrac{b}{a}$ 2) $\dfrac{nb}{a}$ 3) $\dfrac{a}{b}$ 4) $\dfrac{na}{b}$

42. $\underset{h\to 0}{Lim}\dfrac{(2+h)\cos(2+h)-2\cos 2}{h}=$

1) $\cos 2-2\sin 2$ 2) $\cos 2+2\sin 2$
3) $\sin 2-2\cos 2$ 4) $\sin 2+2\cos 2$

43. Let $f(x)=3x^{10}-7x^8+5x^6-21x^3+3x^2-7,$ then the value of $\underset{h\to 0}{Lt}\dfrac{f(1-h)-f(1)}{h^3+3h}$ is

1) $-\dfrac{50}{3}$ 2) $\dfrac{22}{3}$ 3) 13 4) $\dfrac{53}{3}$

Problems using expansions:

44. $\underset{x\to 0}{Lt}\dfrac{\log(1+x)-x}{x^2}=$

1) $\dfrac{1}{2}$ 2) $-\dfrac{1}{2}$ 3) $\dfrac{1}{3}$ 4) $-\dfrac{1}{3}$

Problems using sandwich theorem:

45. $\underset{x\to 0}{Lim}\, x^2\sin\dfrac{\pi}{x}=$

1) 1 2) 0
3) does not exist 4) ∞

46. $\underset{x\to\infty}{Lim}\dfrac{x^2\left(2+\sin^2 x\right)}{x+100}=$

1) 0 2) 1
3) ∞ 4) does not exist

47. $\underset{x\to 0}{Lim}\dfrac{x}{a}\left[\dfrac{b}{x}\right](a\neq 0)$ [where [ ] denotes the G.I.F.) is equal to

1) a 2) b 3) $\dfrac{b}{a}$ 4) $1-\dfrac{b}{a}$

LEVEL-I (H.W)-KEY

1) 1	2) 1	3) 3	4) 4
5) 2	6) 1	7) 3	8) 3
9) 2	10) 1	11) 2	12) 1
13) 1	14) 3	15) 4	16) 4
17) 3	18) 2	19) 4	20) 4
21) 1	22) 2	23) 2	24) 1
25) 1	26) 1	27) 2	28) 4
29) 2	30) 2	31) 2	32) 3
33)3	34) 3	35) 4	36) 3
37) 1	38) 1	39) 2	40) 1
41) 2	42) 1	43) 4	44) 2
45) 2	46) 3	47) 3	

LEVEL-I (H.W)-HINTS

1. Using $\lim_{x\to a}\dfrac{x^p-a^p}{x^q-a^q}=\dfrac{p}{q}a^{p-q}$

2. Using L-Hospital rule $=\lim_{x\to 2}\dfrac{14x-11}{6x-1}=\dfrac{17}{11}$

3. Using L-Hospital rule

$$\lim_{x\to 0}\frac{\frac{1}{m}(1+x)^{\frac{1}{m}-1}+\frac{1}{m}(1-x)^{\frac{1}{m}-1}}{\frac{1}{n}(1+x)^{\frac{1}{n}-1}+\frac{1}{n}(1-x)^{\frac{1}{n}-1}}=\frac{n}{m}$$

4. Using L-Hospital rule
5. Using L-Hospital rule
6. On rationalizing given limit is

$$\lim_{x\to -1}\frac{(x+1)\left(\sqrt{x^2+3}+2\right)}{x^2-1}=-2$$

7. On applying limit $7\leq\lim_{x\to 0}f(x)\leq 7$

$\Rightarrow\lim_{x\to 0}f(x)=7$

8. $\lim_{x\to 1^-}f(x)\neq\lim_{x\to 1^+}f(x)$

9. $\lim_{x\to 0}[\cos x]=\lim_{h\to 0}[\cos(0\pm h)]=\lim_{h\to 0}[\cosh]=0$

$(\because As\ h\to 0,\cosh\to 1)$

$\therefore\lim_{x\to 0}\dfrac{\sin(\cos x)}{1+[\cos x]}=\dfrac{0}{1+0}=0$

10. Use L-H rule

11. Given limit is $\lim_{x\to\frac{\pi}{4}} \dfrac{\tan\left(\frac{\pi}{4}-x\right)}{\frac{\pi}{4}-x}=1$

12. Given limit is $\lim_{x\to 0} \dfrac{(1/4)\sin 3x}{x}=\dfrac{3}{4}$

13. Given limit is $\dfrac{2\sin^2 x\sin 5x}{x^2\sin 3x}=\dfrac{10}{3}$

14. Given limit is

$$\lim_{(x-5)\to 0}\frac{\sin^2(x-5)}{(x-5)^2}\lim_{x\to 5}\frac{\tan(x-5)}{(x+5)}=0$$

15. Given limit $\lim_{x\to 0}\dfrac{2\sin^2\left(\frac{\pi}{180}(x)\right)}{x^2}$

$$=2\left(\frac{\pi}{180}\right)^2$$

16. Given limit is

$$\frac{2\sin^2 x/2}{x\log(1+x)}=\frac{1}{2}\frac{\lim_{x\to 0}\left(\frac{\sin x/2}{x/2}\right)^2}{\lim_{x\to 0}\log(1+x)^{\frac{1}{x}}}=\frac{1}{2}$$

17. $1-\cos\left(\dfrac{x^2}{2}\right)\cong\dfrac{\left(\frac{x^2}{2}\right)^2}{2}$

$$1-\cos\left(\frac{x^2}{4}\right)\cong\frac{\left(\frac{x^2}{4}\right)^2}{2}$$

$$\underset{x\to 0}{Lt}\frac{\frac{x^4}{8}\cdot\frac{x^4}{32}}{x^8}=\frac{1}{256}$$

18. Use LH Rule

19. Use L-Hospital rule

20. Use L-Hospital rule

21. Use L-Hospital rule

22. Use L-Hospital rule

23. $\underset{x\to 0}{Lt}\dfrac{(4^x-1)-(9^x-1)}{x}\times\dfrac{1}{(4^x+9^x)}$

$$=\frac{\underset{x\to 0}{Lt}\left(\frac{4^x-1}{x}\right)-\underset{x\to 0}{Lt}\left(\frac{9^x-1}{x}\right)}{\underset{x\to 0}{Lt}(4^x+9^x)}$$

24. Use L-Hospital rule

25. Rationalising

26. Divide with x in Nr and Dr

27. $\lim_{x\to\infty}\dfrac{\sin\left(\frac{a}{5^x}\right)}{\left(\frac{a}{5^x}\right)}a=a$

28. Divide with $2^{1/n}$

29. Given limit $=\lim_{n\to\infty}\dfrac{\Sigma n.\Sigma n^2}{n.\Sigma n^3}=\dfrac{2}{3}$

30. Given limit

$$=\frac{1}{2}\lim_{n\to\infty}\left(\frac{1}{1}-\frac{1}{3}+\frac{1}{3}-\frac{1}{5}+...+\frac{1}{(2n-1)}-\frac{1}{(2n+1)}\right)$$

$$=\frac{1}{2}$$

31. Given limit $=\lim_{n\to\infty}\dfrac{\sum 2n-1}{\sum 2n}$

32. Divide with $(n+1)!$

33. Given limit is

$$\Rightarrow e^{\lim_{x\to\infty}\left[1+\frac{\lambda}{x}+\frac{\mu}{x^2}-1\right].2x}=e^2$$

$$\Rightarrow e^{\lim_{x\to\infty}\left(\lambda+\frac{\mu}{x}\right).2}=e^2$$

$$\Rightarrow e^{2\lambda}=e^2\Rightarrow\lambda=1,\mu\in R$$

34. Given limit $=e^{\lim_{x\to\infty}(x+b)\left[\frac{x+a}{x+b}-1\right]}$

35. $\lim_{x\to\pi}(1-4\tan x)^{\cot x}=\underset{e^{x\to\pi}}{\lim}\cot x(1-4\tan x-1)=e^{-4}$

36. $\underset{x\to\infty}{Lim}\left(\dfrac{x^2+1}{x^2-1}\right)^{x^2}=e^{\underset{x\to\infty}{Lim}x^2\left(\frac{x^2+1}{x^2-1}-1\right)}=e^2$

37. $\underset{x\to\infty}{Lim}\left(\frac{3x-4}{3x+2}\right)^{\frac{x+1}{3}} = \underset{x\to\infty}{Lim}\, e^{\frac{x+1}{3}\left(\frac{3x-4}{3x+2}-1\right)} = e^{\frac{-2}{3}}$

38. $\lim_{x\to1} x^{\left(\frac{1}{1-x^2}\right)} = e^{\lim_{x\to1}\left(\frac{1}{1-x^2}\right)(x-1)} = e^{\frac{-1}{2}}$

39. Use factorisation in power.

40. $f(x)=x^2+x+1 \Rightarrow \lim_{x\to1}\frac{f(x)-f(1)}{x-1}$

$=\left[f'(x)\right]_{x=1}=3$

41. By L- Hospital rule

Given limit $=\lim_{x\to a}\frac{\frac{1}{2\sqrt{f(x)}}f'(x)}{\frac{1}{2\sqrt{\phi(x)}}.\phi'(x)}=\frac{nb}{a}$

42. By L = Hospital

$\lim_{x\to0}\frac{(2+h)\left[-\sin(2+h)\right]+\cos(2+h)}{1}$

$=-2\sin2+\cos2$

43. Use LH Rule

44. use log (1+x) expansion

45. $\lim_{x\to0}x^2\sin\frac{\pi}{x}=0$

46. Use Sandwich theorem (or) divide by x^2

47. $\frac{b}{x}-1<\left[\frac{b}{x}\right]\le\frac{b}{x}$

LEVEL - II (C.W)

Evaluation of algebra of limits :

1. $\underset{x\to1}{Lim}\frac{\left(\sum_{K=1}^{200}x^K\right)-200}{x-1}=$

1) 5050 2) 1000 3) 2010 4) 20100

2. $\underset{x\to a}{Lt}\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}=$

1) $\frac{2}{\sqrt3}$ 2) $-\frac{1}{\sqrt3}$ 3) $\frac{2}{3\sqrt3}$ 4) $\frac{1}{\sqrt3}$

3. **Let α and β be the roots of $ax^2+bx+c=0$, then**

$\underset{x\to\alpha}{Lt}\frac{1-\cos(ax^2+bx+c)}{(x-\alpha)^2}=$

1) $\frac{a^2(\alpha-\beta)^2}{2}$ 2) $\frac{a^2}{2(\alpha-\beta)^2}$

3) $\frac{a^2}{(\alpha-\beta)^2}$ 4) $-\frac{a^2}{2(\alpha-\beta)^2}$

4. **If** $f(x)=\begin{cases}\sin x, & x\ne n\pi,\ n\in Z\\ 2, & otherwise\end{cases}$

and $g(x)=\begin{cases}x^2+1, & x\ne0,2\\ 4, & x=0\\ 5, & x=2\end{cases}$ **then**

$\lim_{x\to0}g[f(x)]=$

1) 2 2) 4 3) 5 4) 1

5. $\lim_{y\to x}\frac{y^y-x^x}{y-x}=$

1) xy^{x-1} 2) $x^x(1+\log x)$ 3) $y.x^{y-1}$ 4) xy

6. **If** $\lim_{x\to\infty}\left(\frac{x^2-1}{x+1}-ax-b\right)=2$, **then**

1) $a=1$ and $b=-3$ 2) $a=1$ and $b=2$
3) $a=0$ and $b=-1$ 4) $a=2$ and $b=1$

Evaluation of left & right hand limits :

7. **Let $\{x\}$ denote the fractional part of x.**

Then $\lim_{x\to0}\frac{\{x\}}{\tan\{x\}}$ **is equal to**

1) -1 2) 0
3) 1 4) Does not exist

8. $\underset{x\to4^+}{Lt}\frac{x^2-7x+12}{x-[x]}=$ **[where [ ] denotes G.I.F.]**

1) 1 2) 0
3) 2 4) does not exists

9. If $f : R \to R$ defined by

$$f(x)=\begin{cases}\dfrac{x-2}{x^2-3x+2} & if \quad x\in R-\{1,2\}\\ 2 & if \quad x=1\\ 1 & if \quad x=2\end{cases}$$

then $\lim\limits_{x\to2}\dfrac{f(x)-f(2)}{x-2}=$

1) 0 2) -1 3) 1 4) $-1/2$

10. $\underset{x\to o}{Lt}\dfrac{\sqrt{\frac{1}{2}(1-\cos x)}}{x}=$

1) 1 2) -1
3) 0 4) does not exist

Evaluation of trigonometric limits :

11. If $y=\frac{1}{2}\sin^{-1}\left(\frac{2xy}{x^2+y^2}\right)$ and $y<x$ then $\lim\limits_{y\to0}x=$

1) -1 2) 0 3) 1 4) ∞

12. $\underset{x\to\frac{\pi}{4}}{Lt}\dfrac{\sqrt{2}-\cos x-\sin x}{(4x-\pi)^2}=$

1) $\frac{1}{16\sqrt{2}}$ 2) $\frac{1}{32\sqrt{2}}$ 3) $\frac{1}{16}$ 4) $\frac{1}{8}$

13. $\underset{x\to0}{Lt}\dfrac{Secx-1}{x^2(Secx+1)^2}=$

1) 1/8 2) 1/4 3) 2 4) 0

14. $\underset{x\to0}{Lim}\left(\frac{1}{\sin^2x}-\frac{1}{\sinh^2x}\right)=$

1) $\frac{2}{3}$ 2) 0 3) $\frac{1}{3}$ 4) $-\frac{2}{3}$

Evaluation of exponential & logarithmic limits :

15. $\underset{x\to\infty}{Lt}$ x[log(x+1)-logx]=

1) e^2 2) e 3)1 4) 1/e

16. $\lim\limits_{x\to0}\dfrac{27^x-9^x-3^x+1}{\sqrt{2}-\sqrt{1+\cos x}}=$

1) 0 2) $8\sqrt{2}\,(\log3)^2$
3) $8(\log3)^2$ 4) 1

17. $\underset{x\to\infty}{Lt}\, x\left(a^{\frac{1}{x}}-b^{\frac{1}{x}}\right)=$

1) 1 2) $\log_e a/b$
3) $\log_e(ab)$ 4) 0

Limits of the form tends to infinite :

18. The value of

$\underset{n\to\infty}{Lim}\dfrac{1.2+2.3+3.4+...+n.(n+1)}{n^3}$ is

1) 1 2) -1 3) 1/3 4) -1/3

19. If |x|<1, then

$\underset{n\to\infty}{Lt}(1+x)(1+x^2)(1+x^4).....(1+x^{2n})=$

1) $\frac{1}{x}$ 2) $\frac{1}{1+x}$ 3) $\frac{1}{1-x}$ 4) $\frac{1}{x-1}$

20. $\underset{n\to\infty}{Lt}\dfrac{2.3^{n+1}-3.5^{n+1}}{2.3^n+3.5^n}=$

1) 5 2) 1/5 3) -5 4) 0

21. $\lim\limits_{n\to\infty}\frac{1}{n^4}\left[1^2+(1^2+2^2)+...+(1^2+2^2+...+n^2)\right]=$

1) 1/6 2) 1/16 3) 1/12 4) 0

22. $\underset{n\to\infty}{Lt}\dfrac{{}^nP_n}{{}^{n+1}P_{n+1}-{}^nP_n}=$

1) 2 2) -1 3) 0 4) ∞

23. If $a>0$, $\lim\limits_{x\to\infty}\dfrac{[ax+b]}{x}$ is [where [.] denotes G.I.F]

1) 0 2) 1 3) a 4) b

Limit of the form $0^0, 1^\infty, \infty^0$:

24. $\lim\limits_{x\to0}\left(\dfrac{1^x+2^x+3^x+...+n^x}{n}\right)^{\frac{1}{x}}=$

1) $(n!)^n$ 2) $(n!)^{1/n}$ 3) $n!$ 4) $\ln n!$

25. $\underset{x\to0}{Lt}(\cos x)^{\frac{1}{Sinx}}=$

1)1 2) -1 3) 0 4) 3

26. $\lim\limits_{x\to0}\left(\dfrac{\sin x}{x}\right)^{\frac{\sin x}{x-\sin x}}=$

1) e 2) e^2 3) e^3 4) 1/e

27. If $\lim_{x\to 0}\left[1+x\ln(1+b^2)\right]^{\frac{1}{x}}=2b\sin^2\theta, b>0$ and $\theta\in(-\pi,\pi)$ then the value of θ is

1) $\pm\frac{\pi}{6}$ 2) $\pm\frac{\pi}{3}$ 3) $\pm\frac{\pi}{8}$ 4) $\pm\frac{\pi}{2}$

28. $\lim_{x\to 1}(\log_3 3x)^{\log_x 3}=$

1) e^{-1} 2) e 3) -1 4) 1

29. **If p and q are the roots of the quadratic equation** $ax^2+bx+c=0$ **then**

$\lim_{x\to p}\left(1+ax^2+bx+c\right)^{\frac{1}{x-p}}=$

1) $a(p-q)$ 2) $\log\left[a(p-q)\right]$

3) $e^{a(p-q)}$ 4) $e^{a(q-p)}$

30. $\underset{x\to\infty}{Lt}\left(\frac{x^2+5x+3}{x^2+x+2}\right)^x=$

1) e^4 2) e^3 3) e^2 4) 2^4

31. $\underset{x\to\infty}{Lim}\left\{\left(\frac{x}{x+1}\right)^a+\sin\frac{1}{x}\right\}^x$ **is equal to**

1) e^{a-1} 2) e^{1-a} 3) e 4) 0

32. $\lim_{x\to 0}\left\{\tan\left(\frac{\pi}{4}+x\right)\right\}^{\frac{1}{x}}=$

1) e 2) e^2 3) e^{-1} 4) e^{-2}

33. $\lim_{x\to\infty}\left(1+\frac{\lambda}{x}+\frac{\mu}{x^2}\right)^{2x}=e^4$ **then** $\lambda=$ **(** $\mu\in$ **R)**

1) 1 2) 2 3) 3 4) 4

L.H.rule :

34. **A function** $f:R\to R$ **is such that** $f(1)=3$ **and** $f'(1)=6$**. Then**

$\lim_{x\to 0}\left[\frac{f(1+x)}{f(1)}\right]^{1/x}=$

1) 1 2) e^2 3) $e^{1/2}$ 4) e^3

Problems using expansions:

35. **The integer n for which** $\lim_{x\to 0}\frac{(\cos x-1)(\cos x-e^x)}{x^n}$ **is a finite non-zero number is**

1) 1 2) 2 3) 3 4) 4

36. **The values of a,b, and c such that**

$\lim_{x\to 0}\frac{axe^x-b\log(1+x)+cxe^{-x}}{x^2\sin x}=2$

1) $a=3,b=12,c=9$

2) $a=1,b=2,c=4$

3) $a=2,b=10,c=8$

4) $a=3,b=-12,c=-9$

37. $\lim_{x\to 0}\frac{\sin^{-1}x-\sin x}{x^3}$

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ 4) $\frac{1}{5}$

38. $\lim_{x\to 0}\frac{(1+x)^{1/x}-e\left(1-\frac{x}{2}\right)}{(1-\cos x)}$

1) $\frac{1}{2}e$ 2) $\frac{1}{4}e$ 3) $\frac{11}{12}e$ 4) $\frac{1}{12}e$

Problems using sandwich theorem:

39. **If [x] denotes the greatest integer less than or equal to x then**

$\underset{n\to\infty}{Lim}\frac{[x]+[2x]+....+[nx]}{n^2}=$

1) x/2 2) x/3 3) x 4) 0

40. $\lim_{n\to\infty}\frac{\{x\}+\{2x\}+...+\{nx\}}{n^2}$

1) $\frac{1}{2}$ 2) 0 3) -1 4) 2

41. $\underset{x\to 0}{Lt}\left[\frac{\sin|x|}{|x|}\right]$ **Where [.] dentoes the greatest integer function.**

1) 0 2) 1 3) -1 4) does not exist

42. **Let** $f:R\to R$ **be a positive increasing function with** $\lim_{x\to\infty}\frac{f(3x)}{f(x)}=1,$ **then**

$\lim_{x\to\infty}\frac{f(2x)}{f(x)}=$ **[AIEEE 2010]**

1) $\frac{2}{3}$ 2) $\frac{3}{2}$ 3) 3 4) 1

LEVEL-II (C.W)-KEY

1) 4	2) 3	3) 1	4) 4
5) 2	6) 1	7) 4	8)1
9) 2	10) 4	11) 3	12) 1
13) 1	14) 1	15) 3	16) 2
17) 2	18) 3	19) 3	20) 3
21) 3	22) 3	23) 3	24) 2
25) 1	26) 4	27) 4	28) 2
29) 3	30) 1	31) 2	32) 2
33) 2	34) 2	35) 3	36) 1
37) 2	38) 3	39) 1	40) 2
41) 1	42)4		

LEVEL-II (C.W)-HINTS

1. Use L-Hospital rule $\Rightarrow$ Given Limit

$$=\lim_{x\to 1}\left(1+2x+3x^2+..+200.x^{199}\right)$$

2. Using L-Hospital rule given limit is

$$\lim_{x\to a}\frac{\frac{1}{2\sqrt{a+2x}}(2)-\frac{1}{2\sqrt{3x}}(3)}{\frac{1}{2\sqrt{3a+x}}-2\frac{1}{2\sqrt{x}}}=\frac{2}{3\sqrt{3}}$$

3. $ax^2+bx+c=a(x-\alpha)(x-\beta)$

given limit is

$$\lim_{x\to\alpha}\frac{2\sin^2\left(\frac{a(x-\alpha)(x-\beta)}{2}\right)^2}{(x-\alpha)^2}=\frac{a^2(\alpha-\beta)^2}{2}$$

4. Find $(gof)(x)$

5. Using L-Hospital rule,

$$\left(\frac{d}{dy}\right)(y^y)=y^y(1+\log y)$$

$$and\left(\frac{d}{dy}\right)(x^x)=0$$

6. $$\lim_{x\to\infty}\left(\frac{x^2-1-ax^2-ax-bx-b}{x+1}\right)=2$$

$$\Rightarrow\lim_{x\to\infty}\left(\frac{(1-a)x^2-(a+b)x-(1+b)}{x+1}\right)=2$$

$a=1, b+1=-2\Rightarrow b=-3$

7. $$\text{L.H.L}=\lim_{x\to 0^-}\frac{\{x\}}{\tan\{x\}}=\lim_{x\to 0^-}\frac{x-[x]}{\tan(x-[x])}$$

$$=\lim_{x\to 0^-}\frac{x+1}{\tan(x+1)}=\frac{1}{\tan 1}$$

$$\text{R.H.L.}=\lim_{x\to 0^+}\frac{x-[x]}{\tan(x-[x])}=\lim_{x\to 0}\frac{x}{\tan x}=1$$

$\therefore$ L.H.L $\neq$ R.H.L.

8. As $x\to 4^+, [x]=4$

$$\lim_{x\to 4}\frac{(x-3)(x-4)}{x-4}=\lim_{x\to 4}(x-3)=1$$

9. $$\lim_{x\to 2}\frac{f(x)-f(2)}{x-2}=f'(x)$$

10. $$\lim_{x\to 0}\sqrt{\frac{1-\cos x}{2}}=\underset{x\to 0}{Lt}\frac{\sqrt{\sin^2 x/2}}{x}=\underset{x\to 0}{Lt}\frac{|\sin x/2|}{x}$$

does not exist

11. $$y=\frac{1}{2}2\tan^{-1}\left(\frac{y}{x}\right)\Rightarrow x=\frac{\tan y}{y}$$

12. Use L-Hospital rule

13. $$\lim_{x\to 0}\frac{1-\cos x}{x^2}\cdot\left[\frac{1}{\cos x}\cdot\frac{1}{(\sec x+1)^2}\right]=\frac{1}{2}\cdot\frac{1}{4}=\frac{1}{8}$$

14. Use L-Hospital rule

15. $$\lim_{x\to\infty}\log_e\left[1+\frac{a}{x}\right]^x=a$$

16. $$\lim_{x\to 0}\frac{(9.3)^x-9^x-3^x+1}{\sqrt{2}-\sqrt{1+\cos x}}\times\frac{\sqrt{2}+\sqrt{1+\cos x}}{\sqrt{2}+\sqrt{1+\cos x}}$$

$$=\lim_{x\to 0}\frac{(9^x-1)(3^x-1)}{x^2}\frac{x^2}{1-\cos x}\left(\sqrt{2}+\sqrt{1+\cos x}\right)$$

$$\left(\lim_{x\to 0}\frac{a^x-1}{x}=\log a,\lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}\right)$$

17. $$\underset{\frac{1}{x}\to\infty}{Lim}\frac{a^{\frac{1}{x}}-b^{\frac{1}{x}}}{\frac{1}{x}}=\log\left(\frac{a}{b}\right)$$

18. $\underset{n\to\infty}{Lim}\dfrac{\sum_{r=1}^{n} r(r+1)}{n^3}$

$= \underset{n\to\infty}{Lim}\dfrac{\dfrac{n(n+1)(2n+1)}{6}+\dfrac{n(n+1)}{2}}{n^3}$

$= \underset{n\to\infty}{Lim}\dfrac{n(n+1)(n+2)}{3n^3}=\dfrac{1}{3}$

19. $\lim\limits_{x\to\infty} 1+x+x^2+....+x^{2n}=\dfrac{1}{1-x}$

20. $\lim\limits_{x\to\infty}\dfrac{6.3^n-15.5^n}{2.3^n+3.5^n}$

21. $\lim\limits_{n\to\infty}\dfrac{\Sigma\dfrac{n(n+1)(2n+1)}{6}}{n^4}$

22. $\lim\limits_{n\to\infty}\dfrac{n!}{(n+1)!-n!}=\lim\limits_{n\to\infty}\dfrac{1}{n}=0$

23. $\lim\limits_{x\to\infty}\left(a+\dfrac{b}{x}-\dfrac{1}{x}\right)\le\lim\limits_{x\to\infty}\dfrac{[ax+b]}{x}<\lim\limits_{x\to\infty}\left(a+\dfrac{b}{x}\right)$

24. $(1.2.3....n)^{1/n}$

25. $e^{\lim\limits_{x\to0}\frac{\cos x-1}{\sin x}}=e^{\lim\limits_{x\to0}\frac{-\sin x}{\cos x}}$

26. $\lim\limits_{x\to0}\left(\dfrac{\sin x}{x}\right)^{\frac{\sin x}{x-\sin x}}=e^{\lim\limits_{x\to0}\left(\frac{\sin x}{x}-1\right)\frac{\sin x}{x-\sin x}}=1/e$

27. $e^{ln\left(1+b^2\right)}=2b\sin^2\theta$

$\Rightarrow\dfrac{1+b^2}{2b}=\sin^2\theta\Rightarrow\dfrac{1}{b}+b=2\sin^2\theta$

$\Rightarrow\theta=\pm\dfrac{\pi}{2}$

28. $\lim\limits_{x\to1}\left(1+\log_3^x\right)^{\log_x^3}=e$

29. $\lim\limits_{x\to p}\left[1+a(x-p)(x-q)\right]^{\frac{1}{a(x-p)(x-q)}\times a(x-q)}$

$=e^{a(p-q)}$

30. $\lim\limits_{x\to\infty}\left(\dfrac{x^2+5x+3}{x^2+x+2}\right)^x=1^\infty\Rightarrow e^{\lim\limits_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+2}-1\right)x}=e^4$

31. Given limit

$=\underset{x\to\infty}{lim}\left\{1+\left(1+\dfrac{1}{x}\right)^{-a}+\sin\dfrac{1}{x}-1\right\}$

$=\underset{y\to0}{lim}\left\{1+(1+y)^{-a}+\sin y-1\right\}^{1/y}$

where $y=\dfrac{1}{x}$

$=e^{\lim\limits_{y\to0}\frac{(1+y)^{-a}+\sin y-1}{y}}=e^{1-a}$

32. $e^{\lim\limits_{x\to0}\frac{\left(\tan\left(\frac{\pi}{4}+x\right)-1\right)}{x}}=e^{\lim\limits_{x\to0}\frac{\frac{1+\tan x}{1-\tan x}-1}{x}}=e^{\lim\limits_{x\to0}\left(\frac{2\tan x}{1-\tan x}\right)\frac{1}{x}}$

33. $\lambda=2$

34. $\lim\limits_{x\to0}\left(\dfrac{f(1+x)}{f(1)}\right)^{\frac{1}{x}}=e^{\lim\limits_{x\to0}\frac{1}{x}\left\{\frac{f(1+x)}{f(1)}-1\right\}}=e^{\lim\limits_{x\to0}\frac{f'(1)}{f(1)}}=e^2$

35. $\lim\limits_{x\to0}\dfrac{\left(\dfrac{-x^2}{2!}+\dfrac{x^4}{4!}.....\right)\left(-x-x^2+...\right)}{x^n}$

is non-zero if n =3

$ax(1+x+\dfrac{x^2}{2!}+.)-b\left(x-\dfrac{x^2}{2}+\dfrac{x^3}{3}\right)+$

36. $\lim\limits_{x\to0}\dfrac{cx\left(1-\dfrac{x}{1!}+\dfrac{x^2}{2!}+\dfrac{x^3}{3!}\right)}{x^2\left(x-\dfrac{x^3}{3!}+\dfrac{x^5}{5!}\right)}$

coefficient of x & $x^2=0$ and coefficient of $x^3=2$

37. $\lim\limits_{x\to0}\left(x+\dfrac{1^2x^3}{3!}+\dfrac{1^23^2x^5}{5!}....\right)-\left(x-\dfrac{x^3}{3!}+\dfrac{x^5}{5!}\right)$

$\lim\limits_{x\to0}\dfrac{\dfrac{2}{3!}x^3+\dfrac{8}{5!}x^5}{x^3}=\dfrac{1}{3}$

38. $\lim\limits_{x\to0}e\left(1-\dfrac{x}{2}+\dfrac{11}{24}x^2+...\right)-e\left(1-\dfrac{x}{2}\right)$

$=\lim\limits_{x\to0}\dfrac{\dfrac{11}{24}ex^2}{2\sin^2\left(\dfrac{x}{2}\right)}=\dfrac{11e}{12}\left(\dfrac{\dfrac{x}{2}}{\sin\dfrac{x}{2}}\right)=\dfrac{11}{12}e$

39. By using Sandwitch theorem

40. $0 \le \{x\} < 1$

$0 \le \{2x\} < 1$

$\frac{0}{n^2} \le \frac{\{x\}+\{2x\}+.....+\{nx\}}{n^2} < \frac{n}{n^2}$

$\lim_{n\to\infty}\frac{0}{n^2} \le \lim_{n\to\infty}\frac{\{x\}+\{2x\}+.....+\{nx\}}{n^2} \le \lim_{n\to\infty}\frac{1}{n}$

$\lim_{n\to\infty} f_1(x) = \lim_{n\to\infty}\frac{0}{n} = 0$

$\lim_{n\to\infty} f_2(x) = \lim_{n\to\infty}\frac{1}{n} = 0$

$\lim_{n\to\infty}\frac{\{x\}+\{2x\}+.....+\{nx\}}{n^2} = 0$

41. If $0 < |x| < \frac{\pi}{2}$ then $|\sin x| < |x| < |\tan x|$

$\therefore \frac{\sin|x|}{|x|} < 1$

$\left[\frac{\sin|x|}{|x|}\right] = 0$

42. By using Sandwich theorem

LEVEL - II (H.W)

Evaluation of algebra of limits :

1. $\underset{x\to 0}{Lt}\frac{(1+x)^3-1-3x}{(1+x)^2-1-2x} =$

1) 2 2) 3 3) 5 4) -2

2. If f(x) = $\frac{4-7x}{7x+4}$, $\underset{x\to 0}{Lt}$ f(x) = l and $\underset{x\to\infty}{Lt} f(x) = m$ the quadratic equation having roots as $\frac{1}{l}$ and $\frac{1}{m}$ is

1) $x^2-1=0$ 2) $x^2+1=0$

3) 1/2 4) $x^3-1=0$

3. $\underset{x\to 2}{Lim}\frac{\sqrt{x+7}-3\sqrt{2x-3}}{\sqrt[3]{x+6}-2\sqrt[3]{3x-5}} =$

1) $\frac{34}{23}$ 2) $\frac{23}{17}$ 3) $\frac{7}{23}$ 4) $\frac{23}{7}$

4. $\underset{x\to 1}{Lim}\left[\frac{x^3+2x^2+x+1}{x^2+2x+3}\right]^{\frac{1-\cos(x-1)}{(x-1)^2}} =$

1) e 2) $e^{1/2}$ 3) 1 4) $\left(\frac{5}{6}\right)^{1/2}$

5. $\underset{x\to 0}{Lim}\frac{x\sqrt{y^2-(y-x)^2}}{\left\{\sqrt{(8xy-4x^2)}+\sqrt{8xy}\right\}^3} =$

1) $\frac{1}{4y}$ 2) $\frac{1}{2}$ 3) $\frac{1}{2\sqrt{2}}$ 4) $\frac{1}{128y}$

6. $\underset{x\to\infty}{Lt}\frac{x^5}{5^x} =$ ____

1) 0 2) 1 3) ∞ 4) does not exist

7. $\lim_{x\to 8}\frac{\sqrt{1+\sqrt{1+x}}-2}{x-8} =$ **[EAMCET - 2011]**

1) $\frac{3}{2}$ 2) $\frac{1}{4}$ 3) $\frac{1}{24}$ 4) $\frac{1}{5}$

Evaluation of left & right hand limits :

8. $\lim_{x\to 2}\left(\frac{\sqrt{1-\cos\{2(x-2)\}}}{x-2}\right)$ **[AIEEE 2011]**

1) equals $\sqrt{2}$ 2) equals $-\sqrt{2}$

3) equals $\frac{1}{\sqrt{2}}$ 4) does not exist

9. If $l_1 = \lim_{x\to 2^+}(x+[x]), l_2 = \lim_{x\to 2^-}(2x-[x])$ and $l_3 = \lim_{x\to\frac{\pi}{2}}\frac{\cos x}{x-\frac{\pi}{2}}$ then [where [ ] denotes G.I.F.]

1) $l_1 < l_2 < l_3$ 2) $l_2 < l_3 < l_1$

3) $l_3 < l_2 < l_1$ 4) $l_1 < l_3 < l_2$

Evaluation of trigonometric limits :

10. If $a = \min\left\{x^2+4x+5, x \in R\right\}$ and $b = \lim\limits_{\theta \to 0} \frac{1-\cos 2\theta}{\theta^2}$ then the value of $\sum\limits_{r=0}^{n} a^r b^{n-r} =$

1) $\frac{2^{n+1}-1}{4.2^n}$ 2) $2^{n+1}-1$

3) $\frac{2^{n+1}-1}{3.2^n}$ 4) 2^n-1

11. $\lim\limits_{x \to 0} \frac{\tan x - \sin x}{x^2} =$ **[EAMCET - 2010]**

1. 0 2. 1 3. $\frac{1}{2}$ 4. $-\frac{1}{2}$

12. $\lim\limits_{x \to 0} \frac{(1-\cos 2x)(3+\cos x)}{x \tan 4x} =$ **[JEE M 2013]**

1) $\frac{-1}{4}$ 2) $\frac{1}{2}$ 3) 1 4) 2

13. $\lim\limits_{x \to 0} \frac{\tan^3 x - \sin^3 x}{x^5} =$ **[EAMCET - 2013]**

1) $\frac{5}{2}$ **2)** $\frac{3}{2}$ **3)** $\frac{3}{5}$ **4)** $\frac{2}{5}$

14. $\underset{x \to 0}{Lt} \frac{\mathrm{Sin2x}+2\mathrm{Sin}^2\mathrm{x}-2\mathrm{Sinx}}{\mathrm{Cosx}-\mathrm{Cos}^2\mathrm{x}} =$

1) 0 2) 1 3) $\frac{2}{3}$ 4) 4

15 $\underset{x \to 0}{Lt} \frac{3\sin \mathrm{x} - \sin 3\mathrm{x}}{\mathrm{x}.\tan^2 2\mathrm{x}} =$

1) 0 2) 1 3) 2 4) 4

16. $\underset{x \to 0}{Lim} \frac{8}{x^8}\left(1-\cos\frac{x^2}{2}-\cos\frac{x^2}{4}+\cos\frac{x^2}{2}.\cos\frac{x^2}{4}\right) =$

1) $\frac{1}{16}$ 2) $\frac{1}{15}$ 3) $\frac{1}{32}$ 4) 1

17. **Arrange the following limits in the ascending order.**

1) $\underset{x \to 0}{Lim} \frac{\tan^4 x - \sin^4 x}{x^6}$

2) $\underset{x \to 0}{Lim} \frac{\tan^8 x - \sin^8 x}{x^5 \tan x^5}$

3) $\underset{x \to 0}{Lim} \frac{\tan^3 x - \sin^3 x}{x \sin^4 x}$

4) $\underset{x \to 0}{Lim} \frac{\tan^5 x - \sin^5 x}{x^2.\sinh^3 x.\tan^2 x}$

1) 1, 2, 3, 4 2) 3, 1, 4, 2

3) 1, 2, 4, 3 4) 2, 1, 3, 4

18. **The vaue of θ, is**

$$\lim_{\theta \to 0} \frac{\cos^2\left\{1-\cos^2\left(1-\cos^2....\left(\cos^2\left\{1-\cos^2\theta\right\}\right)\right)\right\}}{\sin\left(\frac{\pi(\sqrt{\theta+4}-2}{\theta}\right)}$$

1) $\frac{\sqrt{2}}{4}$ 2) $\sqrt{2}$ 3) 1 4) 2

19. $\underset{x \to \pi}{Lt} \frac{\sqrt{2+\cos x}-1}{(\pi-x)^2} =$ ____

1) 0 2) $\frac{1}{4}$ 3) $\frac{1}{2}$ 4) 2

20. $If \lim\limits_{x \to 0} \frac{\{(a-n)nx - \tan x\}\sin nx}{x^2} = 0,$ **where n is a non-zero real number, then 'a' =**

1) 0 2) $\frac{n+1}{n}$ 3) n 4) $n+\frac{1}{n}$

21. $\underset{h \to 0}{Lim} 2\left[\frac{\sqrt{3}\sin\left(\frac{\pi}{6}+h\right)-\cos\left(\frac{\pi}{6}+h\right)}{\sqrt{3}.h\left(\sqrt{3}\cosh-\sinh\right)}\right] =$

1) $\frac{-2}{\sqrt{3}}$ 2) $\frac{-4}{3}$ 3) $\frac{2}{3}$ 4) $\frac{4}{3}$

Evaluation of exponential & logarithemic limits :

22. The value of $\lim_{x\to a}\frac{\log(x-a)}{\log(e^x-e^a)}$ is

1) 1 2) -1 3) 0 4) 2

23. Arrange the following limits in the ascending order

1) $\lim_{x\to\infty}\left(\frac{1+x}{2+x}\right)^{x+2}$ 2) $\lim_{x\to 0}(1+2x)^{3/x}$

3) $\lim_{\theta\to 0}\frac{\sin\theta}{2\theta}$ 4) $\lim_{x\to 0}\frac{\log_e(1+x)}{x}$

1) 1, 2, 3, 4 2) 1, 3, 4, 2
3) 1, 4, 3, 2 4) 3, 4, 1, 2

24. $\lim_{x\to 0}\frac{(4^x-1)^3}{\sin\left(\frac{x}{4}\right)\log_e\left(1+\frac{x^2}{3}\right)}=$

1) $(\log_e 4)^3$ 2) $\log_e 4$

3) $12(\log_e 4)^3$ 4) $5(\log_e 4)^3$

Limits of the form tends to infinite :

25. $\lim_{x\to 0}\frac{e^{\frac{1}{x}}-1}{e^{\frac{1}{x}}+1}=$

1) 1 2) -1 3) 0 4) does not exists

26. $\lim_{n\to\infty}\frac{1^3+2^3+3^3+....+n^3}{3n^4+5n^3+6}=$

1) 1/3 2) 1/5 3) 1/6 4)1/12

27. $\lim_{x\to\infty}\frac{3.\sqrt{x}+5\sin^2 x-10.\log x}{5.\sqrt{x}+7\cos^2 x+100.\log x}=$

1) $\frac{3}{5}$ 2) $\frac{5}{3}$ 3) 15 4) $\frac{1}{15}$

28. $\lim_{n\to\infty}\frac{\left(\sqrt{n^2+1}+n\right)^2}{\sqrt[3]{n^6+1}}=$

1) 1 2) 1/2 3) 1/3 4) 4

29. $\lim_{x\to\infty}\frac{(2+x)^{20}(4+x)^3}{(2-x)^{23}}=$

1) – 1 2) 1 3) 6 4) 2

30. $\lim_{n\to\infty}\cos\left(\pi\sqrt{n^2+n}\right)$ in equal to

1)0 2)1
3) 2 4) does not exist

31. $\lim_{n\to\infty}\left(\frac{1}{5}\right)^{\log_{\sqrt{5}}\left(\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+...to\ n\ terms\right)}$ equals

1) 2 2) 4 3) 8 4) 0

32. $\lim_{n\to\infty}\frac{1^4+2^4+3^4+...+n^4}{n^5}$

$-\lim_{n\to\infty}\frac{1^3+2^3+3^3+...+n^3}{n^5}$ is

1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{3}{5}$ 4) $\frac{4}{5}$

33. If a > 1, then the value of $\lim_{x\to\infty}\frac{a^{\sqrt{x}}-a^{1/\sqrt{x}}}{a^{\sqrt{x}}+a^{1/\sqrt{x}}}$ is

1) 1 2) 0 3) 2 4) 3

Limit of the form $0^0, 1^\infty, \infty^0$:

34. for $x>0; \lim_{x\to 0}\left((\sin x)^{\frac{1}{x}}+\left(\frac{1}{x}\right)^{\sin x}\right)=$

1) 0 2) -1 3) 1 4) 2

35. $\lim_{x\to 0}\left(\frac{1+\tan x}{1+\sin x}\right)^{cosec x}$ is equal to

1) $\frac{1}{e}$ 2) e 3) e^2 4)1

36. $\lim_{x\to\infty}\left(\frac{x+6}{x+1}\right)^{x+4}=$ [EAMCET - 2012]

1. e^4 2. e^6 3. e^5 4. e

37. $\lim_{x\to\infty}\left(\frac{x+5}{x+2}\right)^{x+3}=$ [EAMCET - 2009]

1) e 2) e^2 3) e^3 4) e^5

L.H.rule :

38. **If** $f^1(0)=3$, **then**

$$\lim_{x\to 0}\frac{x^2}{f(x^2)-6f(4x^2)+5f(7x^2)}=$$

1) $\frac{1}{36}$ 2) $-\frac{1}{36}$ 3) $\frac{1}{34}$ 4) $\frac{1}{106}$

Problems using expansions:

39. $\lim_{x\to 0}\frac{(1+x)^{1/x}-e}{x}=$

1) 1 2) e/2 3) –e/2 4)2/e

Problems using sandwich theorem:

40. **If [x] denotes the greatest integer less than or equal to x then**

$$\lim_{n\to\infty}\frac{1}{n^3}\left\{\left[1^2x\right]+\left[2^2x\right]+\left[3^2x\right]+......+\left[n^2x\right]\right\}=$$

1) $\frac{x}{2}$ 2) $\frac{x}{3}$ 3) $\frac{x}{6}$ 4) 0

LEVEL-II (H.W)-KEY

1) 2	2) 1	3) 1	4) 4
5)4	6)1	7)3	8)4
9) 3	10) 2	11)1	12)4
13)2	14) 4	15) 2	16) 3
17) 2	18)2	19)2	20)4
21) 4	22) 1	23) 2	24)3
25) 4	26) 4	27) 1	28) 4
29) 1	30) 1	31)2	32)1
33) 1	34)3	35)4	36)3
37)3	38) 1	39)3	40) 2

LEVEL-II (H.W)-HINTS

1. Using L-Hospital rule given limit is

$$\lim_{x\to 0}\frac{6(1+x)}{2}=3$$

2. $l=1, m=-1$,

$$\frac{1}{l}=1,\frac{1}{m}=-1\Rightarrow x^2-1=0$$

3. Using L-Hospital rule

4. $\left(\frac{5}{6}\right)^{1/2}$

5. Take common higher power of x *in both numerator and denominater*

6. $\lim_{x\to\infty}\frac{x^5}{5^x}=\lim_{x\to\infty}\frac{x^5}{e^{x\log 5}}$

$$\lim_{x\to\infty}\frac{x^5}{1+mx+\frac{m^2x^2}{2!}+....}$$ where $m=\log 5$

$$\lim_{x\to\infty}\frac{1}{\frac{1}{x^5}+\frac{m}{x^4}+\frac{m^2}{2x^3}+\frac{m^3}{6x^2}+\frac{m^4}{24x}+\frac{m^5}{120}+\frac{m^6x}{720}+...}$$

$$=\frac{1}{\infty}=0$$

7. Use L-Hospital's rule

8. $\sqrt{1-\cos 2\theta}=\sqrt{2}\left|\sin\theta\right|$

9. Difinition of $[x]$ and L hospital rule

10. $a=\frac{4ac-b^2}{4a}=\frac{4.5.1-16}{4}=1$

$$b=\frac{2\sin 2\theta}{2\theta}=2$$

$$\sum_{r=0}^{n}a^rb^{n-r}=b^n+ab^{n-1}+a^2b^{n-2}+.....+a^n$$

$$=\frac{2^{n+1}-1}{2-1}$$

11. Take $\tan x$ common and simplify

12. $\lim_{x\to 0}\frac{1-\cos 2x}{x^2}\times\frac{3+\cos x}{1}\times\frac{x}{\tan 4x}$

13. $\frac{n}{2}$

14. Given limit is

$$\frac{2\sin x(\cos x-1)+2(1-\cos^2 x)}{x\cos x(1-\cos x)}=4$$

15. $\lim_{x\to 0}\frac{4\sin^3 x}{x\tan^2 2x}=\frac{4}{4}=1$

16. Given limit is

$$\lim_{x\to 0}\frac{8}{x^8}\left(1-\cos\frac{x^2}{2}\right)\left(1-\cos\frac{x^2}{4}\right)$$

$$=32\frac{1}{16}\frac{1}{64}=\frac{1}{32}$$

17. $\lim_{x\to 0}\frac{\tan^n ax-\sin^n ax}{x^{n+2}}=\frac{n}{2}a^{n+2}$

18. Given limit

$$=\lim_{\theta\to 0}\frac{1}{\sin\left[\frac{\pi}{\sqrt{\theta+4}+2}\right]}$$

$$=\frac{1}{\sin\frac{\pi}{4}}=\sqrt{2}$$

19. $\lim_{x\to \pi}\frac{\sqrt{2+\cos x}-1}{(\pi-x)^2}$

$$=\lim_{y\to 0}\frac{\sqrt{2+\cos(\pi-y)}-1}{y^2}$$

$$=\lim_{y\to 0}\frac{\sqrt{2-\cos y}-1}{y^2}$$

$$=\lim_{y\to 0}\frac{2-\cos y-1}{y^2}\times\frac{1}{\sqrt{2-\cos y}+1}$$

$$=\lim_{y\to 0}\frac{2\sin^2\frac{y}{2}}{\left(\frac{y}{2}\right)^2}\times\frac{1}{4}\times\frac{1}{\sqrt{2-\cos y}+1}$$

$$=2\times 1\times\frac{1}{4}\times\frac{1}{1+1}=\frac{1}{4}$$

20. $\lim_{x\to 0}\left\{(a-n)n-\frac{\tan x}{x}\right\}\frac{\sin nx}{x}=0$,

$$\Rightarrow n\left[(a-n)n-1\right]=0$$

$$\Rightarrow (a-n)n=1\Rightarrow a=n+\frac{1}{n}$$

21. Using L.H Rule

22. $\lim_{x\to a}\frac{\log(x-a)}{\log(e^x-e^a)}$

$$=\lim_{x\to a}\frac{e^x}{(x-a)e^x+e^x}=\frac{e^a}{e^a}=1$$

23. Using standard formulae

24. Divide with x^3 and simplify

25. L.H.L$=\lim_{h\to 0}\frac{e^{\frac{-1}{h}}-1}{e^{\frac{-1}{h}}+1}=-1$

R.H.L$=\lim_{h\to 0}\frac{e^{\frac{1}{h}}-1}{e^{\frac{1}{h}}+1}=1$

26. Given limit $=\lim_{n\to\infty}\frac{\frac{n^2(n+1)^2}{4}}{3n^4+5n^3+6}=\frac{1}{12}$

27. Divide with $\sqrt{x}$

28. $\frac{(1+1)^2}{\sqrt[3]{1+0}}=4$

29. Divide with x^{23} and applying limit

30. $\sqrt{n^2+n}=n\sqrt{1+\frac{1}{n}}$

$$=n\left(1+\frac{1}{n}\right)^{1/2}$$

$$=n\left(1+\frac{1}{2n}\right)$$

$$=\left(n+\frac{1}{2}\right)$$

$$\lim_{n\to\infty}\cos\left(\pi\left(n+\frac{1}{2}\right)\right)=\lim_{n\to\infty}\sin(n\pi)=0$$

31. $\lim_{n\to\infty}5^{-2\log_5\left[\frac{1}{4}\frac{\left(1-\left(\frac{1}{2}\right)^n\right)}{1-1/2}\right]}$

$$\lim_{n\to\infty} 5^{\log_5\left[\frac{1}{4}\left(1-\left(\frac{1}{2}\right)^n\right)\times\frac{2}{1}\right]^{-2}}$$

$$\lim_{n\to\infty}\left(\frac{1}{4}\times\frac{2}{1}\right)^{-2} = (2^{-1})^{-2} = 2^2 = 4$$

$$\left\{\begin{array}{l} S_n \text{ of } G.P = \dfrac{a(1-r^n)}{1-r} \text{ if } r<1 \\ a^{\log_a x} = x \end{array}\right.$$

32. $\displaystyle lim_{n\to\infty}\frac{n(n+1)(2n+1)(3n^2+3n-1)}{30n^5}$

$\displaystyle = lim_{n\to\infty}\frac{n^2(n+1)^2}{4n^5} = \frac{6}{30} - 0 = \frac{1}{5}$

33. $a^{-\infty} = 0$

34. $x\to 0 : (\sin x)^{\frac{1}{x}} \to 0$ and $\displaystyle\lim_{x\to 0}\left(\frac{1}{x}\right)^{\sin x}$ is ∞^0 form

35. 1^{∞} form

36. $e^{6-1} = e^5$

37. $e^{5-2} = e^3$

38. By L-Hospital rule

39. $(1+x)^{1/x} = e^{1-\frac{x}{2}+\frac{x^2}{3}-\frac{x^3}{4}+......\infty}$

40. Sandwich theorem

LEVEL - III

1. $\displaystyle\lim_{x\to 0}\frac{e^{x^3}-1-x^3}{\sin^6(2x)} =$

1) $\frac{1}{128}$ 2) $\frac{2}{127}$ 3) $\frac{1}{126}$ 4) $\frac{1}{125}$

2. $\displaystyle\lim_{x\to\frac{\pi}{2}}\frac{1-(\sin x)^{\sin x}}{\cos^2 x} =$

1) 2 2) 1 3) 1/2 4) 1/4

3. If $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$ **then** $\displaystyle\lim_{x\to 0}\frac{f'(x)}{x} =$

1) 1 2) – 1 3) 2 4) – 2

4. $\displaystyle\lim_{x\to 0}\left(\frac{\log_{\sec\left(\frac{x}{2}\right)}\cos x}{\log_{\sec x}\cos\left(x/2\right)}\right) =$

1) 14 2) 15 3) 16 4) 17

5. $\displaystyle\lim_{x\to 0}\left[\frac{100\tan x.\sin x}{x^2}\right]$ **where** $[.]$ **represents greatest integer function is**

1) 99 2) 100 3) 0 4) 98

6. If $\{x\}$ **denotes fractional part of x, then** $\displaystyle\lim_{x\to 1}\frac{x\sin\{x\}}{x-1} =$

1) 0 2) -1 3) 1 4) does not exist

7. The graph of the function $y = f(x)$ **has a unique tangent at the point** $(e^a, 0)$ **through which the graph passes then** $\displaystyle\lim_{x\to e^a}\frac{\log_e\{1+7f(x)\}-\sin f(x)}{3f(x)}$ **is**

1) 1 2) 2 3) 0 4) -1

8. If $[.]$ **denotes the greatest integer function, then** $\displaystyle\lim_{x\to\frac{\pi}{2}}\left[\frac{x-\frac{\pi}{2}}{\cos x}\right] =$

1) 1 2) -1 3) 2 4) -2

9. If $[.]$ **denotes the greatest integer function then** $\displaystyle\lim_{x\to 0}\left[\frac{x^2}{\tan x.\sin x}\right] =$

1) 0 2) 1
3) -1 4) does not exist

10. $\displaystyle\lim_{n\to\infty}\frac{1}{n^4}\sum_{r=1}^{n} r(r+2)(r+4) =$

1) $\frac{3}{4}$ 2) 0 3) $\frac{1}{8}$ 4) $\frac{1}{4}$

11. $\lim_{n\to\infty}\left[\frac{7}{10}+\frac{29}{10^2}+\frac{133}{10^3}+........+\frac{5^n+2^n}{10^n}\right]=$

1) 3/4 2) 2 3) 5/4 4) 1/2

12. **Suppose f(n+1)** $=\frac{1}{2}\left\{f(n)+\frac{9}{f(n)}\right\}, n\in N.$ **If** $f(n)>0, \forall n\in N,$ **then** $\lim_{n\to\infty}(f(n))=$

1) 3^{-1} 2) -3^{-1} 3) 3 4) -3

13. $\lim_{x\to 0}\left[1^{1/\sin^2 x}+2^{1/\sin^2 x}+....+n^{1/\sin^2 x}\right]^{\sin^2 x}=$

1) ∞ 2) 0 3) $\frac{n+1}{2}$ 4) n

14. **The value of**

$\lim_{x\to 0}\left(\frac{e^{[\log(2^x-1)]^x}-(2^x-1)^x\sin x}{e^{x\log x}}\right)^{1/x}$ **is equal to**

1) e 2) $\frac{1}{e}\log 2$ 3) $e\log 2$ 4) $\log 2$

15. $\lim_{n\to\infty}(n^4+n^3+A_1n^2+A_2n+A_3)^{1/2}-(n^4+n^3+B_1n^2+B_2n+B_3)^{1/2}$ **equals**

1) $\frac{A_1-B_1}{2}$ 2) $\frac{A_1+B_3}{2}$

3) $\frac{B_3-A_1}{2}$ 4) $\frac{A_1-B_3}{2}$

16. $\lim_{n\to\infty}\frac{1.n^2+2(n-1)^2+3(n-2)^2+...+n.1^2}{1^3+2^3+...+n^3}$ **equals**

1) $\frac{8}{3}$ 2) $\frac{4}{3}$ 3) $\frac{2}{3}$ 4) $\frac{1}{3}$

17. $\lim_{n\to\infty}\frac{1-2+3-4+5-6+.....-2n}{\sqrt{n^2+1}+\sqrt{4n^2-1}}=$

1) $\frac{1}{3}$ 2) $-\frac{1}{3}$ 3) $-\frac{1}{5}$ 4) $\frac{1}{5}$

18. **If** $\lim_{x\to 0}\left(x^{-3}\sin 3x+ax^{-2}+b\right)$ **exists and is equal to zero,then the value of** $a+2b=$

1) 3 2) 4 3) 0 4) 6

19. **The graph of** $y=f(x)$ **has unique tangent at the point** $(a,0)$ **through which the graph passes. Then** $\lim_{x\to a}\frac{\log\left[1+6f(x)\right]}{3f(x)}=$

1) 0 2) 1 3) 2 4) ∞

20. **The value of**

$\lim_{x\to 0}\left(1-\frac{1}{2^x}\right)\left(\frac{1}{\sqrt{\tan x+4}-2}\right)$ **is**

1) $\log 16$ 2) does not exist

3) $3\log 2$ 4) $6\log 2$

21. **The value of**

$\lim_{x\to 0}\left\{\left[\frac{100x}{\sin x}\right]+\left[\frac{99\sin x}{x}\right]\right\}$**, where** [.] **represents the greatest integer function, is**

1) 199 2) 198 3) 0 4) 1

22. $\lim_{x\to 0^-}\frac{[x]+[x^2]+[x^3]+...+[x^{2n+1}]+n+1}{1+[x^2]+[x]+2x}$

$n\in N$ **is equal to**

1) $n+1$ 2) n 3) 1 4) 0

23. **If** [.] **denotes the greatest integer function, then**

$\lim_{x\to 0}\frac{\tan\left(\left[-2\pi^2\right]x^2\right)-x^2\tan\left(\left[-2\pi^2\right]\right)}{\sin^2 x}=$

1) $-20+\tan 20$ 2) $20+\tan 20$

3) 20 4) $\tan 20$

24. $\lim_{n\to\infty}\frac{2^3-1^3}{2^3+1^3}.\frac{3^3-1^3}{3^3+1^3}....\frac{n^3-1}{n^3+1}$ **equals**

1) $\frac{1}{3}$ 2) $\frac{2}{3}$ 3) 1 4) $\frac{3}{2}$

25. $\lim_{n\to\infty}{}^nC_r\left(\frac{m}{n}\right)^r\left(1-\frac{m}{n}\right)^{n-r}$ **equals**

1) $e^{-m}m^r$ 2) $\frac{m^r}{r!}$ 3) $\frac{m^re^{-m}}{r!}$ 4) $\frac{e^{-r}r^m}{m!}$

26. **The value of** $\lim_{x\to 0}\left\{\sin^2\left(\frac{\pi}{2-ax}\right)\right\}^{\sec^2\frac{\pi}{2-bx}}$ **is**

equal to

1) $e^{-a/b}$ 2) e^{-a^2/b^2} 3) $a^{2a/b}$ 4) $e^{4a/b}$

27. $\lim_{x\to\infty}\left[\frac{1^2}{1-x^3}+\frac{3}{1+x^2}+\frac{5^2}{1-x^3}+\frac{7}{1+x^2}+...\right]=$

1) $\frac{-5}{6}$ 2) $\frac{-10}{3}$ 3) $\frac{5}{6}$ 4) $\frac{10}{3}$

28. Evaluate $\lim_{n\to\infty}\left(1+\frac{1}{a_1}\right)\left(1+\frac{1}{a_2}\right)....\left(1+\frac{1}{a_n}\right)$

where $a_1=1$ and $a_n=n(1+a_{n-1})\forall n\geq 2$

1) $\frac{1}{e}$ 2) $\frac{1}{e^2}$ 3) e 4) 1

29. $\lim_{n\to\infty} n^{-n^2}\left\{(n+1)\left(n+\frac{1}{2}\right)\left(n+\frac{1}{2^2}\right)......\left(n+\frac{1}{2^{n-1}}\right)\right\}^n=$

1) $\frac{1}{2e^2}$ 2) $\frac{1}{e}$ 3) e^2 4) $\frac{3}{2}e^2$

30. Evaluate

$\lim_{n\to\infty}\left\{\frac{1}{2}\tan\frac{x}{2}+\frac{1}{2^2}\tan\frac{x}{2^2}+...+\frac{1}{2^n}\tan\frac{x}{2^n}\right\}$

1) $x\tan\frac{x}{2}$ 2) $\frac{1}{x}\cot\frac{x}{2}$

3) $\frac{x-\cot x}{2}$ 4) $\frac{1}{x}-\cot x$

31. $\lim_{x\to-\pi}\frac{|x+\pi|}{\sin x}=$

1) 1 2) -1

3) π 4) does not exit

32. If $\ell(x)$ is least integer not lessthan x and $g(x)$ is the greatest integer not greaterthan x then

$\lim_{x\to e+\pi}(\ell(x)+g(x))=$

1) 1 2) 9 3) 11 4)13

33. If $0<P<1$ then $\lim_{n\to\infty}\frac{n^P\sin^2(n!)}{n+1}=$

1) 0 2) 1 3) ∞ 4) $\frac{4}{3}$

34. $\lim_{x\to a^+}\frac{\{x\}\sin(x-a)}{(x-a)^2}=$ **where** $\{x\}$ **denotes fractional part of x and $a\in N$**

1) 0 2) 1 3) a 4) 5

35. $\lim_{x\to 0}\left\{\left[\frac{a\sin x}{x}\right]+\left[\frac{b\tan x}{x}\right]\right\}=$

$a,b\in N$, **[where [ ] denotes G.I.F.]**

1) $a+b$ 2) $a+b-1$ 3) 0 4) $\frac{a+b}{2}$

36. $\lim_{x\to 0}\frac{\cos(\sin x)-\cos x}{x^4}=$

1) $\frac{1}{6}$ 2) $\frac{1}{5}$ 3) $\frac{1}{4}$ 4) $\frac{1}{2}$

37. $\lim_{x\to 0}\frac{1-\sin[\cos x]}{[x]-[\sin x]}=$ **(where $[x]$ denotes greatest integral part of x)**

1) 0 2) 1 3) 2 4) ∞

38. $\lim_{x\to -1}\frac{1}{\sqrt{|x|-\{-x\}}}=$ **(where $\{x\}$ denotes fractional part of x)**

1) does not exist 2) 1

3) ∞ 4) $\frac{1}{2}$

39. $\lim_{x\to 0}\left[\frac{\sin(sgn(x))}{(sgn(x))}\right]=$

(where $[x]$ denotes integral part of x)

1) 0 2) 1 3) -1 4) does not exist

40. If $f(x)=|x-1|-[x]$ (where [x] is greatest integer less than or equal to x) then.

1) $f(1^+)=-1; f(1^-)=0$

2) $f(1^+)=0=f(1^-)$

3) $\lim_{x\to 1}f(x)$ **exits**

4) Cannot say any thing.

41. $\lim_{x\to -2^-}\frac{ae^{\frac{1}{|x+2|}}-1}{2-e^{\frac{1}{|x+2|}}}=\lim_{x\to -2^+}\sin\left(\frac{x^4-16}{x^5+32}\right)$ then a is

1) $\sin\left(\frac{3}{5}\right)$ 2) 2 3) $\sin\left(\frac{2}{5}\right)$ 4) $\sin\left(\frac{1}{5}\right)$

42. If $\{x\}$ denotes fractional part of x then

$\lim_{x\to 1}\frac{x\sin\{x\}}{x-1}$

1) 0 2) -1 3) 1 4) does not exist

43. $\lim_{n\to\infty}\sum_{x=1}^{20}\cos^{2n}(x-10)=$

1) 0 2) 1 3)19 4) 20

44. **If $f(x)$ is a real number in [0, 1], then the value of the function**

$f(x)=\lim_{m\to\infty}\lim_{n\to\infty}\left(1+\cos^{2m}(n!\pi x)\right)$ **is given by**

1) 2 or 1 according as x is a rational or irrational

2) 2 or 1 according as x is irrational or rational

3) 1 for all x 4) 2 for all x

45. $\lim_{x\to\infty}\sec^{-1}\left(\frac{x}{x+1}\right)$ **is equal to**

1) 0 2) π

3) $\frac{\pi}{2}$ 4) Does not exist

46. **If $n\in N$ then $\lim_{x\to n}(-1)^{[x]}$, where $[x]$ denotes the greatest integer less than or equal to x, is equal to**

1) $(-1)^n$ 2) $(-1)^{n-1}$

3) 0 4) does not exist

47. $\lim_{x\to-1}\frac{\cos 2-\cos 2x}{x^2-|x|}$ **is equal to**

1) 2 2) sin 2 3) 2 sin 2 4) 0

48. **The value of** $\lim_{x\to\infty}\left(1+\frac{1}{x^n}\right)^x, n>0$ **is**

1) 1, $if\ n<1$ 2) 1, $if\ n>1$

3) e, $if\ n>1$ 4) e, $if\ n<1$

49. $\lim_{x\to 0}\left(1+x\left(1+\frac{f(x)}{kx^2}\right)\right)^{1/x}=e^3$ **and** $f(4)=64$ **then K has value**

1) 1 2) 2 3) 4 4) 5

LEVEL-III-KEY

1)1	2)3	3)4	4)3
5)1	6)4	7)2	8)4
9)1	10)4	11)3	12)3
13)4	14)2	15)1	16)4
17)2	18)4	19)3	20)1
21)2	22)4	23)1	24)2
25)3	26)2	27)2	28)3
29)3	30)4	31)4	32)3
33)1	34)2	35)2	36)1
37)4	38)1	39)1	40)1
41)3	42)4	43)2	44)1
45)4	46)4	47)3	48)2
49)2			

LEVEL-III-HINTS

1. $e^{x^3}=1+x^3+\frac{(x^3)^2}{2!}+\frac{(x^3)^3}{3!}+...\infty$

$\Rightarrow e^{x^3}-1-x^3=x^6\left[\frac{1}{2}+\frac{x^3}{6}+...+\infty\right]$

2. Use L hospital rule

3. $f(x)=-x^2\cos x+x^2\tan x$

4. Use L hospital rule

5. We have $0<\frac{\tan x.\sin x}{x^2}<1$ in the nbd

$x=0\Rightarrow 0<\frac{100\tan x.\sin x}{x^2}<100$

6. $\lim_{x\to1^-}\frac{x\sin\{x\}}{x-1}=\lim=\frac{x\sin x}{x-1}=-\infty$

$\lim_{x\to1^-}\frac{x\sin\{x\}}{x-1}=\lim_{x\to1}=\frac{x\sin(x-1)}{x-1}=1$

7. $f(e^a)=0$

Given limit

$$=\lim_{x\to e^a}\frac{\frac{1}{1+7.f(x)}7.f'(x)-\cos[f(x)].f'(x)}{3.f'(x)}$$

$$=\frac{\frac{7}{1+7f(e^a)}-\cos[f(e^a)]}{3}$$

$= \frac{7-1}{3} = 2$

8. Given limit

$= \lim_{\theta \to 0}\left[\frac{\theta}{\cos\left(\frac{\pi}{2}+\theta\right)}\right]$ where

$x - \frac{\pi}{2} = \theta$

9. $\sin x < x < \tan x$ in the nbd of 0

$\therefore 0 < \frac{x^2}{\tan x \sin x} < 1$

10. $\lim_{n\to\infty}\frac{1}{n^4}\left[\sum n^3 + 6\sum n^2 + 8\sum n\right] = \frac{1}{4}$

11. $\lim_{n\to\infty}\left(\left(\frac{5}{10}\right)^n + \left(\frac{2}{10}\right)^n\right)$

12. $\lim_{n\to\infty} f(n+1) = \lim_{n\to\infty} f(n) = k$

$k = \frac{1}{2}\left[k + \frac{9}{k}\right] \Rightarrow k^2 = 9, k = 3$

13. Given limit

$= \lim_{x\to 0} n.\left[\left(\frac{1}{n}\right)^{\frac{1}{\sin^2 x}} + \left(\frac{2}{n}\right)^{\frac{1}{\sin^2 x}} + ... + \left(\frac{n-1}{n}\right)^{\frac{1}{\sin^2 x}} + 1\right]$

$= n(0+0+...+0+1) = n$

14. $\lim_{x\to 0}\left[\frac{(2^x-1)^x - (2^x-1)^x \sin x}{x^x}\right]^{1/x}$

$= \lim_{x\to 0}\frac{2^x-1}{x}\lim_{x\to 0}(1-\sin x)^{1/x}$

$= \lim_{x\to 0}\frac{2^x-1}{x}.e^{\lim_{x\to 0}\frac{1}{x}(1-\sin x - 1)} = \frac{\log 2}{e}$

15. On rationalising, given limit $= \frac{A_1 - B_1}{2}$

16. Given expansion $= \sum_{r=1}^{n}\frac{[r.(n-r+1)^2]}{\sum_{r=1}^{n} r^3}$

$= \sum_{r=1}^{n}\frac{(n+1)^2 r - 2(n+1)r^2 + r^3}{\frac{n^2(n+1)^2}{4}}$

$= \frac{(n+1)^2\frac{n(n+1)}{2} - 2(n+1)\frac{n(n+1)(2n+1)}{6} + \frac{n^2(n+1)^2}{4}}{\frac{n^2(n+1)^2}{4}}$

Given limit $= 2 + 1 - \frac{8}{3} = \frac{1}{3}$

17. Given $\lim_{n\to\infty}\frac{n^2-(n+n^2)}{\sqrt{n^2+1}+\sqrt{4n^2-1}}$ and divide with 'n'

18. $\lim_{x\to 0}\frac{\sin 3x}{x^3} + \frac{a}{x^2} + b$

$= \lim_{x\to 0}\frac{\sin 3x + ax + bx^3}{x^3}$

Use L'Hospitals rule

19. Use L hospitals rule

20. $\lim_{x\to 0}\left(\frac{2^x-1}{2^x.x}\right)\left(\frac{\sqrt{\tan x + 4}+2}{\frac{\tan x + 4 - 4}{x}}\right) = 4\log 2 = \log 16$

21. $\forall x > 0; \frac{x}{\sin x} > 1$ and $\forall x < 0; \frac{\sin x}{x} < 1$

$\frac{x}{\sin x} > 1$ and $\frac{\sin x}{x} < 1$

$\Rightarrow \frac{100x}{\sin x} > 100$ and $\frac{99\sin x}{x} < 99$

22. $\left[x^{2n+1}\right] = -1$ and $\left[x^{2n}\right] = 0$ for $n = 0,1,2,3,....$ Given limit

$= \lim_{x\to 0^-}\frac{(-1)+0+(-1)+0+...+0+(-1)+n+1}{1+0-1+2x} = 0$

23. $\left[-2\pi^2\right] = -20$

24. Given limit

$$=\lim_{n\to\infty}\left(\frac{1.7}{3.3}\right)\left(\frac{2.13}{4.7}\right)\left(\frac{3.21}{5.13}\right)\left(\frac{4.31}{6.21}\right)...\left(\frac{(n-1)(n^2+n+1)}{(n+1)(n^2-n+1)}\right)$$

$$=\lim_{n\to\infty}\frac{2}{3}.\frac{(n^2+n+1)}{(n^2-n+1)}=\frac{2}{3}$$

25. $\lim_{n\to\infty} {}^nC_r\left(\frac{m}{n}\right)^r\left(1-\frac{m}{n}\right)^{n-r}$

$$=\lim_{n\to\infty}\frac{n(n-1)(n-2)...(n-r+1)}{r!}.\frac{m^r}{n^r}\left[\left(1-\frac{m}{n}\right)^{-\frac{n}{m}}\right]^{-\frac{(n-r)\times m}{n}}$$

$$=\lim_{n\to\infty}\frac{n(n-1)(n-2)...(n-r+1)}{n^r\quad r!}(m^r)$$

$$=\lim_{n\to\infty}\left[\left(1-\frac{m}{n}\right)^{-\frac{n}{m}}\right]^{\left(1-\frac{r}{n}\right)(-m)}$$

$$=\lim_{n\to\infty}\frac{\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)...\left(1-\left(\frac{r-1}{n}\right)\right)}{r!}m^r.e^{(1-0)(-m)}$$

$$=\frac{(1-0)(1-0)...(1-0)}{r!}m^r\ e^{-m}$$

$$=\frac{e^{-m}\ m^r}{r!}$$

26. Given limit is of the form 1^∞

$\therefore$ Given limit $=e^{\lim_{x\to0}\sec^2\left(\frac{\pi}{2-bx}\right)\left(\sin^2\left(\frac{\pi}{2-ax}\right)-1\right)}$

$$=e^{\lim_{x\to0}}\frac{\cos^2\left(\frac{\pi}{2-ax}\right)}{\cos^2\left(\frac{\pi}{2-bx}\right)}$$

Use L-Hospital rule $=e^{\lim_{x\to0}}\dfrac{\sin\left(\frac{2\pi}{2-ax}\right)}{\sin\left(\frac{2\pi}{2-bx}\right)}.\dfrac{a}{b}$

Again use L-Hospital rule $=e^{\frac{-a^2}{b^2}}$

27. $\lim_{x\to\infty}\left[\dfrac{\sum_{k=1}^{x}(4k-3)^2}{1-x^3}+\sum_{k=1}^{x}\dfrac{4k-1}{1+x^2}\right]$

$$=\frac{16}{-3}+\frac{4}{2}=\frac{-16}{3}+2=\frac{-10}{3}$$

28.

$$L=\lim_{n\to\infty}\left(1+\frac{1}{a_1}\right)\left(1+\frac{1}{a_2}\right)....\left(1+\frac{1}{a_n}\right)$$

$$=\lim_{n\to\infty}\left(\frac{a_1+1}{a_1}\right)\left(\frac{a_2+1}{a_2}\right)....\left(\frac{a_n+1}{a_n}\right)$$

$$=\lim_{n\to\infty}\frac{\left(\frac{a_2}{2}\right)\left(\frac{a_3}{3}\right)....\left(\frac{a_{n+1}}{n+1}\right)}{a_1a_2....a_n}$$

($\because$ use $\frac{a_n}{n}=1+a_{n-1}\forall n\geq 2$ & $a_1=1$ given)

$$\lim_{n\to\infty}\frac{a_{n+1}}{a_1\lfloor n+1}=\lim_{n\to\infty}\frac{a_n+1}{\lfloor n+1}$$

$$\lim_{n\to\infty}\frac{(n+1)[1+a_n]}{\lfloor(n+1)}=\lim_{n\to\infty}\frac{1}{n!}+\frac{a_n}{n!}$$

$$\lim_{n\to\infty}\left(\frac{1}{n!}+\frac{1}{(n-1)!}+\frac{a_{n-1}}{(n-1)!}\right)$$

proceeding in a similar manner weight

$$L=\lim_{n\to\infty}\left(\frac{1}{n!}+\frac{1}{(n-1)!}+.....\frac{1}{2!}+\frac{1}{1!}+1\right)=e$$

29. $\lim_{n\to\infty}n^{-n^2}\left\{(n+1)\left(n+\frac{1}{2}\right)\left(n+\frac{1}{2^2}\right)....\left(n+\frac{1}{2^{n-1}}\right)\right\}^n$

$$=\lim_{n\to\infty}\left\{\frac{(n+1)\left(n+\frac{1}{2}\right)....\left(n+\frac{1}{2^{n-1}}\right)}{n^n}\right\}^n$$

$\left[\because n^{n^2}=\left(n^n\right)^n\right]$

$$=\lim_{n\to\infty}\left\{\left(1+\frac{1}{n}\right)^n\left[\left(1+\frac{1}{2n}\right)^{2n}\right]^{\frac{1}{2}}\left[\left(1+\frac{1}{4n}\right)^{4n}\right]^{\frac{1}{4}}...\right.$$

$$\left.\left[\left(1+\frac{1}{2^{n-1}n}\right)^{2^{n-1}n}\right]^{\frac{1}{2^{n-1}}}\right\}$$

$$=e.e^{\frac{1}{2}}e^{\frac{1}{4}}.....=e^{1+\frac{1}{2}+\frac{1}{4}+...}=e^2$$

30 $\lim_{n\to\infty}\left\{\frac{1}{2}\tan\frac{x}{2}+\frac{1}{2^2}\tan\frac{x}{2^2}+...+\frac{1}{2^n}\tan\frac{x}{2^n}\right\}$

$$=\lim_{n\to\infty}\left\{-\cot x+\left(\cot x+\frac{1}{2}\tan\frac{x}{2}\right)\right.$$

$$\left.+\frac{1}{2^2}\tan\frac{x}{2^2}+...+\frac{1}{2^n}\tan\frac{x}{2^n}\right\}$$

$$=\lim_{n\to\infty}\left\{-\cot x+\left(\frac{1}{2}\cot\frac{x}{2}+\frac{1}{2^2}\tan\frac{x}{2^2}\right)\right.$$

$$\left.+.....+\frac{1}{2^n}\tan\frac{x}{2^n}\right\}$$

$\left(\because \cot x+\frac{1}{2}\tan\frac{x}{2}=\frac{1}{2}\cot\frac{x}{2}\right)$

proceeding like this we get

$$=\lim_{n\to\infty}\left\{-\cot x+\frac{1}{2^n}\cot\frac{x}{2^n}\right\}$$

$$=-\cot x+\lim_{n\to\infty}\left(\frac{x/2^n}{\tan\frac{x}{2^n}}.\frac{1}{x}\right)=-\cot x+\frac{1}{x}$$

31. $\lim_{x\to-\pi^+}\frac{|x+\pi|}{\sin x}=\lim_{n\to 0}\frac{|-\pi+h+\pi|}{\sin(-\pi+h)}$

$$=\lim_{h\to 0}\frac{|h|}{-\sin(\pi-h)}$$

$$=\lim_{h\to 0}\frac{h}{-\sinh}=-1$$

$$\lim_{x\to-\pi^-}\frac{|x+\pi|}{\sin x}=\lim_{n\to 0}\frac{|-\pi-h+\pi|}{\sin(-\pi-h)}$$

$$=\lim_{n\to 0}\frac{|-h|}{-\sin(\pi+h)}=\lim_{n\to 0}\frac{h}{\sinh}=1$$

$\therefore$ L.H.L $\neq$ R.H.L

$\Rightarrow$ limit does not exist

32. Let $f(x)=\ell(x)+g(x)$

$e\approx 2.708,\ \pi\approx 3.142,\ e+\pi\approx 5.85$

$$\lim_{x\to(e+\pi)^+}f(x)=\lim_{h\to 0}(e+\pi+h)$$

$$=\lim_{h\to 0}\{\ell(e+\pi+h)+g(e+\pi+h)\}$$

$$\lim_{h\to 0}(6+5)=11$$

use $5<e<\pi\pm h<6$

$$\lim_{x\to(e+\pi)^-}f(x)=\lim_{h\to 0}(e+\pi-h)$$

$$\lim_{h\to 0}\{\ell(e+\pi-h)+g(e+\pi-h)\}$$

$$=\lim_{h\to 0}(6+5)=11$$

$$\therefore \lim_{x\to e+\pi}f(x)=11$$

33. $\lim_{n\to\infty}\frac{n^p\sin^2(n!)}{n+1}=\lim_{n\to\infty}\frac{n^P\sin^2(n!)}{n\left(1+\frac{1}{n}\right)}$

$$=\lim_{n\to\infty}\frac{\sin^2(n!)}{n^{1-p}\left(1+\frac{1}{n}\right)}=0$$

$(\because$ as $n\to\infty$ $\frac{1}{n^{1-p}}\to 0$ &

$$\sin^2 n!\le 1\&\left(1+\frac{1}{n}\right)\to 1$$

34. $f(x)=\frac{\{x\}\sin(x-a)}{(x-a)^2}=\frac{(x-[x])\sin(x-a)}{(x-a)^2}$

$$\lim_{x\to a^+}f(x)=\lim_{n\to 0}\frac{(a+h-[a+h])\sin(a+h-a)}{(a+h-a)^2}$$

$$\lim_{h\to 0}\frac{(a+h-a)\sinh}{h^2}=1$$

35. $\frac{\sin x}{x}<1\Rightarrow\frac{a\sin x}{x}<a$ but close to $a\Rightarrow\left[\frac{a\sin x}{x}\right]=a-1$

$\frac{\tan x}{x}>1\Rightarrow\frac{b\tan x}{x}>b$ but close to b $\Rightarrow\left[\frac{b\tan x}{x}\right]=b$

$$\therefore\lim_{x\to 0}\left\{\left[\frac{a\sin x}{x}\right]+\left[\frac{b\tan x}{x}\right]\right\}=$$

$$a-1+b=a+b-1$$

36. $\lim_{x\to 0}\frac{\cos(\sin x)-\cos x}{x^4}$

$$\lim_{x\to 0}-\frac{2}{x^4}\sin\left(\frac{\sin x+x}{2}\right)\sin\left(\frac{\sin x-x}{2}\right)$$

$$=\lim_{x\to 0}-2\frac{\sin\left(\frac{\sin x+x}{2}\right)}{\left(\frac{\sin x+x}{2}\right)}\cdot\frac{\sin\left(\frac{\sin x-x}{2}\right)}{\left(\frac{\sin x-x}{2}\right)}\times$$

$$\left(\frac{\sin x+x}{x}\right)\left(\frac{\sin x-x}{x^3}\right)\frac{1}{4}$$

$$=\frac{-2}{4}\times 1\times 1\times\underset{x\to 0}{Lt}\left(\frac{\sin x}{x}+1\right)\times$$

$$\lim_{x\to 0}\left(\frac{x-\frac{x^3}{3!}+\frac{x^5}{5!}......x}{x^3}\right)$$

$$=\frac{-1}{2}\times(1+1)\times\left(-\frac{1}{6}\right)=\frac{1}{6}$$

37. $\lim_{x\to 0^-}\frac{1-\sin[\cos x]}{[x]-[\sin x]}=\lim_{n\to 0}\frac{1-\sin[\cos(-h)]}{[-h]-[\sin(-h)]}$

$$\lim_{n\to 0}\frac{1-\sin 0}{-1-(-1)}=\frac{1}{0}=\infty$$

$$\lim_{x\to 0^+}\frac{1-\sin[\cos x]}{[x]-[\sin x]}=\lim_{n\to 0}\frac{1-\sin[\cosh]}{[h]-[\sinh]}$$

$$=\lim_{h\to 0}\frac{1-\sin 0}{0-0}=\frac{1}{0}=\infty$$

$$\therefore\lim_{x\to 0}\frac{1-\sin[\cos x]}{[x]-[\sin x]}=\infty$$

use $\cosh<1\Rightarrow[\cosh]=0$
$h>0,h\to 0,[h]=0$
$[-h]=-1$
$[\sinh]=0$
$[-\sinh]=-1$

$$=2\times 1^3\times\frac{0}{8\times 1}=0$$

38. L.H.L=

$$\lim_{x\to -1^-}\frac{1}{\sqrt{|x|-\{-x\}}}=\lim_{x\to -1}\frac{1}{\sqrt{-x-(x+2)}}$$

$$=\lim_{x\to -1}\frac{1}{\sqrt{-2x-2}}=\infty$$

R.H.L=

$$\lim_{x\to -1^+}\frac{1}{\sqrt{|x|-\{-x\}}}=\lim_{x\to -1}\frac{1}{\sqrt{-x-(x+1)}}$$

$$\lim_{x\to -1}\frac{1}{\sqrt{-2x-1}}=1$$

L.H.L $\neq$ R.H.L $\Rightarrow$ limit does not exist

39. $\lim_{x\to 0^+}\left[\frac{\sin(\text{sgn}(x))}{\text{sgn}(x)}\right]=\lim_{x\to 0}\left[\frac{\sin 1}{1}\right]=0$

$$\lim_{x\to 0^-}\left[\frac{\sin(\text{sgn}(x))}{\text{sgn}(x)}\right]=\lim_{x\to 0}\left(\frac{\sin(-1)}{-1}\right)$$

$$\lim_{x\to 0}[\sin 1]=0$$

$$\therefore \lim_{x\to 0}\left[\frac{\sin(\text{sgn}(x))}{\text{sgn}(x)}\right]=0$$

use $\text{sgn}(x)=1, x>0$
$=-1, x<0$
$=0, x=0$

40. Find $f(1^+)$ and $f(1^-)$ and verify

41. $\lim_{x\to -2^-}\frac{ae^{\frac{1}{|x+2|}}-1}{2-e^{\frac{1}{|x+2|}}}=\lim_{x\to -2^-}\frac{a-e^{-\frac{1}{|x+2|}}}{2e^{-\frac{1}{|x+2|}}-1}=-a$

$$\lim_{x\to -2^+}\sin\left(\frac{x^4-16}{x^5+32}\right)$$

$$\sin\left(\lim_{x\to -2^+}\frac{\frac{x^4-(-2)^4}{x-(-2)}}{\frac{x^5-(-2)^5}{x-(-2)}}\right)=\sin\left(\frac{-2}{5}\right)$$

$$-a=\sin\left(-\frac{2}{5}\right)$$

$$a=\sin\left(\frac{2}{5}\right)$$

42. $\lim_{x\to 1^-}\frac{x\sin\{x\}}{x-1}=\lim_{x\to 1^-}\frac{x\sin x}{x-1}=-\infty$

$$=\lim_{x\to 1^+}\frac{x\sin\{x\}}{x-1}=\lim_{x\to 1^+}\frac{x\sin(x-1)}{x-1}=1$$

43. $\lim_{n\to\infty}\cos^{2n}x=1$ where $x=m\pi, m\in I$

$=0$ where $x\neq m\pi, m\in I$

Here for $x=10$, $\lim_{n\to\infty}\cos^{2n}(x-10)=1$ and in all other cases it is zero

$$\therefore \lim_{n\to\infty}\sum_{x=1}^{20}\cos^{2n}(x-10)=1$$

44. $con(n!\pi x)=\pm 1$ for x is rational

$\cos^{2m}(n!\pi x)\to 0$ as $m\to\infty$ for x is irrational

45. $x<x+1$ for $x>0\Rightarrow \frac{x}{x+1}<1$

$\sec^{-1}\left(\frac{x}{x+1}\right)$ is not defined

46. $L.H.L=\lim_{h\to 0}(-1)^{[n-h]}=(-1)^{n-1}$

$R.H.L=\lim_{h\to 0}(-1)^{[n+h]}=(-1)^{n}$

47. $\lim_{x\to -1}\frac{2\sin(1+x)\sin(x-1)}{x^2+x}$

$$2\lim_{x\to -1}\frac{\sin(x+1)}{x+1}\times\frac{\sin(x-1)}{x}$$

48. $\lim_{x\to\infty}\left(1+\frac{1}{x^n}\right)^x=e^{\lim_{x\to\infty}\frac{x}{x^n}}=e^{\lim_{x\to\infty}x^{1-n}}=\begin{cases}e^0=1, & if\, n>1\\ e, & if\, n=1\\ \infty, & if\, n<1\end{cases}$

49. $\lim_{x\to 0}\left(1+x\left(1+\frac{f(x)}{kx^2}\right)\right)^{1/x}=e^{1+\frac{f(x)}{kx^2}}$

LEVEL - IV

1. **I :** $\lim_{n\to\infty}\frac{1+3+6+......+\frac{n(n+1)}{2}}{n^3}=\frac{1}{6}$

II : $\lim_{n\to\infty}\frac{1.1!+2.2!+3.3.!+.......+n.n!}{(n+1)!}=1$

1) Only I is true
2) Only II is true
3) Both I and II are true
4) Neither I nor II is true

2. **I :** $\lim_{\theta\to 0}\left(\left[\frac{n.\sin\theta}{\theta}\right]+\left[\frac{n\tan\theta}{\theta}\right]\right)=$ **even integer where [.] denotes the greatest integer function.**

II : $\lim_{x\to 0}\frac{ae^x-b}{x}=2$ **then a =2, b =2**

1) Only I is true
2) Only II is true
3) Both I and II are true
4) Neither I nor II is true

3. **Arrange the following limits in the descending order**

a) $\lim_{x\to 3-}\frac{3-x}{|x-3|}$

b) $\lim_{x\to 0}\frac{(1+x)^{\frac{1}{8}}-(1-x)^{\frac{1}{8}}}{x}$

c) $\lim_{x\to 0}\frac{\tan x-\sin x}{x^3}$

d) $\lim_{x\to\infty}\left\{\sqrt{x^2+6x+7}+x\right\}$

1) b,c,d,a 2) d,a,c,b
3) d,c, b, a 4) b,c, a, d

4. **Assertion(A) : If |x| < 1 then**

$\lim_{n\to\infty}(1+x)(1+x^2)(1+x^4)......(1+x^{2n})=\frac{1}{1-x}$

Reason (R): $(1+x)(1+x^2)......\left(1+x^{2n}\right)=\frac{1-x^{4n}}{1-x}$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true and R is not the correct explanation of A
3) A is true but R is false
4) R is true but A is false

5. **Assertion(A) :** $\lim_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+2}\right)^x=e^4$

Reason(R) : $\lim_{x\to\infty}(1+x)^{1/x}=e$

1) Both A and R are true and R is the correct
explanation of A
2) Both A and R are true and R is not the correct explanation of A
3) A is true but R is false
4) R is true but A is false

6. **List - I** **List - II**

List - I	List - II
a) $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$	1) $e^{\frac{-1}{6}}$
b) $\lim_{x\to 0}\left(\frac{\tan x}{x}\right)^{\frac{1}{x^2}}$	2) $e^{\frac{1}{3}}$
c) $\lim_{x\to 0}(\cos x)^{\frac{1}{\tan x}}$	3) $e^{\frac{2}{\pi}}$
d) $\lim_{x\to a}\left(2-\frac{x}{a}\right)^{Tan\left(\frac{\pi x}{2a}\right)}$	4) 1

Then the correct match for List - I from List - II

	a	b	c	d		a	b	c	d
1)	1	4	3	2	2)	1	2	3	4
3)	2	1	4	3	4)	1	2	4	3

7. **If m, n are +ve integers and** $a_0, b_0 \neq 0$ **are non-zero real numbers**

$s=\lim_{x\to\infty}\frac{a_0x^m+a_1x^{m-1}+.........+a_{m-1}x+a_m}{b_0x^n+b_1x^{n-1}+...........+b_{n-1}x+b_n}$ **then**

List - I	List - II
a) If m = n	1) s= $\frac{a_0}{b_0}$
b) If m < n	2) s= ∞
c) If m > n	3) s=0

Then the correct match for List - I from List - II

	a	b	c		a	b	c
1)	1	2	3	2)	1	3	2
3)	2	1	3	4)	1	3	3

8. **Match the following :**

List-I	List-II
A) $\lim_{x\to\infty}\left(\frac{a^{1/x}+b^{1/x}+c^{1/x}}{3}\right)^{\frac{1}{x}} =$	1) 1/2
B) $\lim_{x\to 5}\frac{x^2-9x+20}{x-[x]} =$	2) –1
C) $\lim_{x\to 0}\frac{1-e^{1/x^2}}{1+e^{1/x^2}} =$	3) does not exist
D) $\lim_{n\to\infty}\frac{2.3^n+5.2^n}{4.3^n-7.2^n} =$	4) $(abc)^{1/3}$

	A	B	C	D
1)	1	2	3	4
2)	4	3	3	1
3)	4	3	2	1
4)	4	3	1	2

9. **I :** $\lim_{n\to\infty}\frac{n!}{(n+1)!-n!}=0$

II : $\lim_{x\to 0}\sqrt{\frac{x-\sin^2 x}{x+\cos x}}=1$

1) Only I is true 2) Only II is true
3) Both I and II are true
4) Neither I nor II is true

10. **I : If a > 0 then** $\lim_{x\to\infty}\frac{[ax+b]}{x}=a$ **, [Where $[x]$ denotes greatest integral part of x].**

II : $\lim_{x\to\frac{\pi}{2}}[\sin x]=0$

[Where $[x]$ denotes greatest integral part of x].

1) Only I is true
2) Only II is true
3) Both I and II are true
4) Neither I nor II is true

11. **I :** $\lim_{x\to 0}\left(\frac{\tan[e^2]x^2-\tan[-e^2]x^2}{\sin^2 x}\right)=15,$

[Where $[x]$ denotes greatest integral part of x].

II : $\lim_{x\to 0}\left(\frac{1^x+2^x+3^x+......+n^x}{n}\right)^{\frac{1}{x}}=(n!)^{1/n}$

1) Only I is true 2) Only II is true
3) Both I and II are true
4) Neither I nor II is true

LEVEL-IV-KEY

1) 3	2) 2	3) 2	4) 1
5) 3	6) 4	7) 2	8) 3
9) 3	10) 3	11) 3	

LEVEL-IV-HINTS

1. 1) $\frac{\sum\frac{n^2+n}{2}}{n^3}=\frac{1}{6}$

2) $\frac{(n+1)!-1}{(n+1)!}$

2. Using step function definition
3. By simplifying
4. A is true, R is true $R\Rightarrow A$
5. A is true , R is false
6. 1^∞ form
7. Divide Nr with x^m
Divide Dr with x^m
8. Using standard formula
9. 1) Divide with $n!$
2) Divide with x
10. $ax+b-1\le[ax+b]<ax+b$

$$a+\frac{b}{x}-\frac{1}{x}\le\frac{[ax+b]}{x}<a+\frac{b}{x}$$

$$\lim_{x\to\infty}\frac{[ax+b]}{x}=a$$

11. Standard formula

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